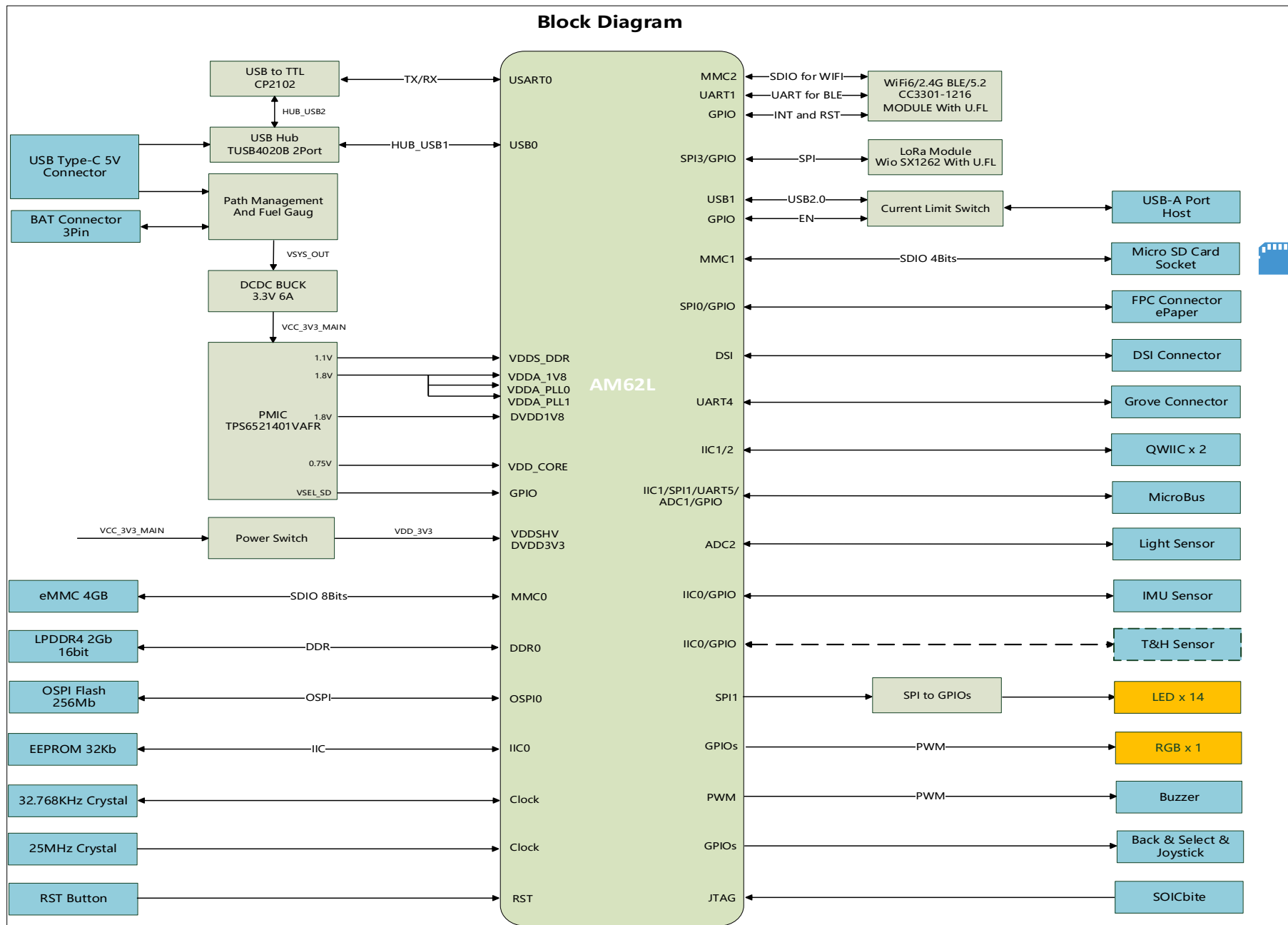


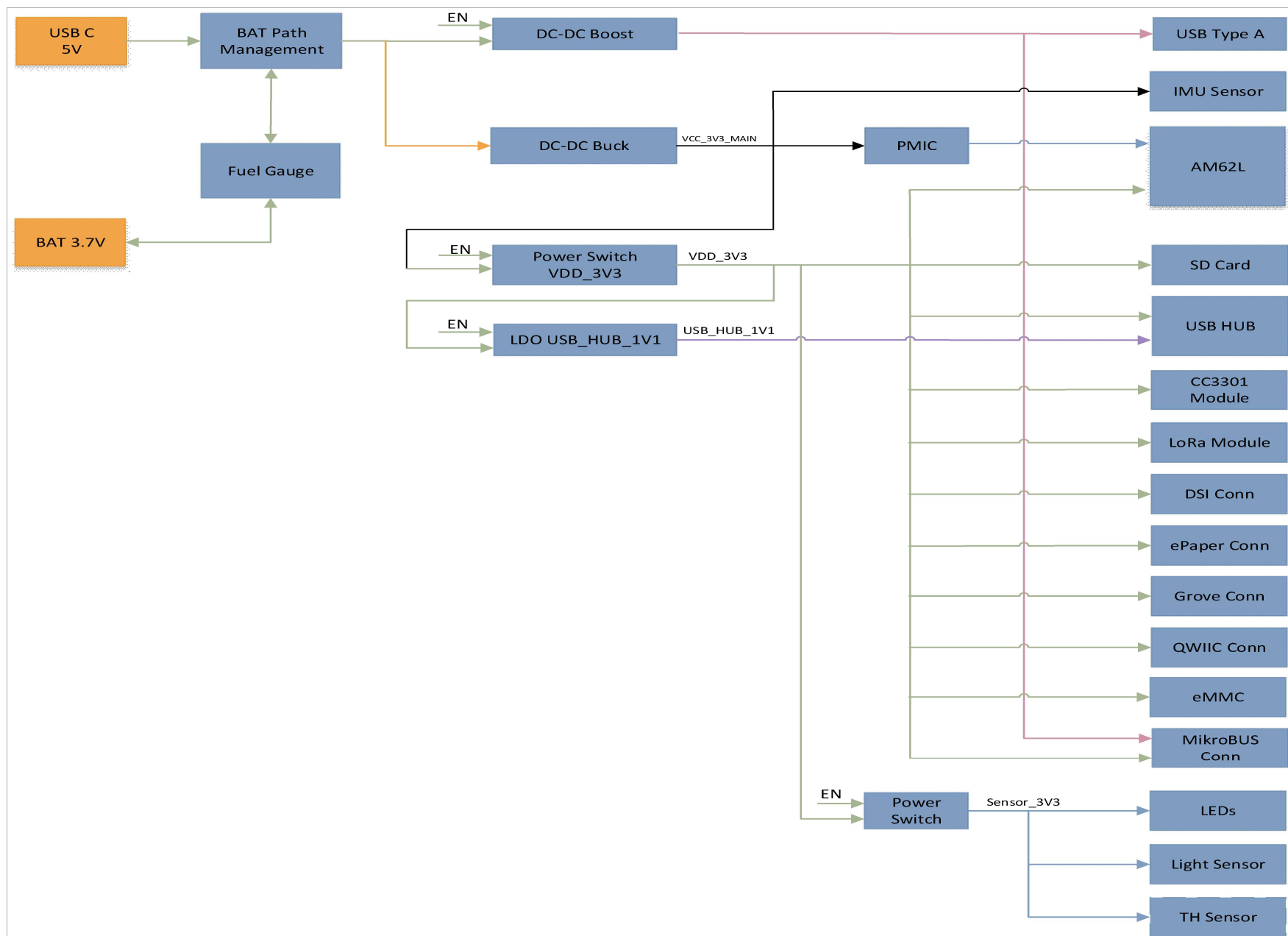
Schematic:BeagleBadge AM62L

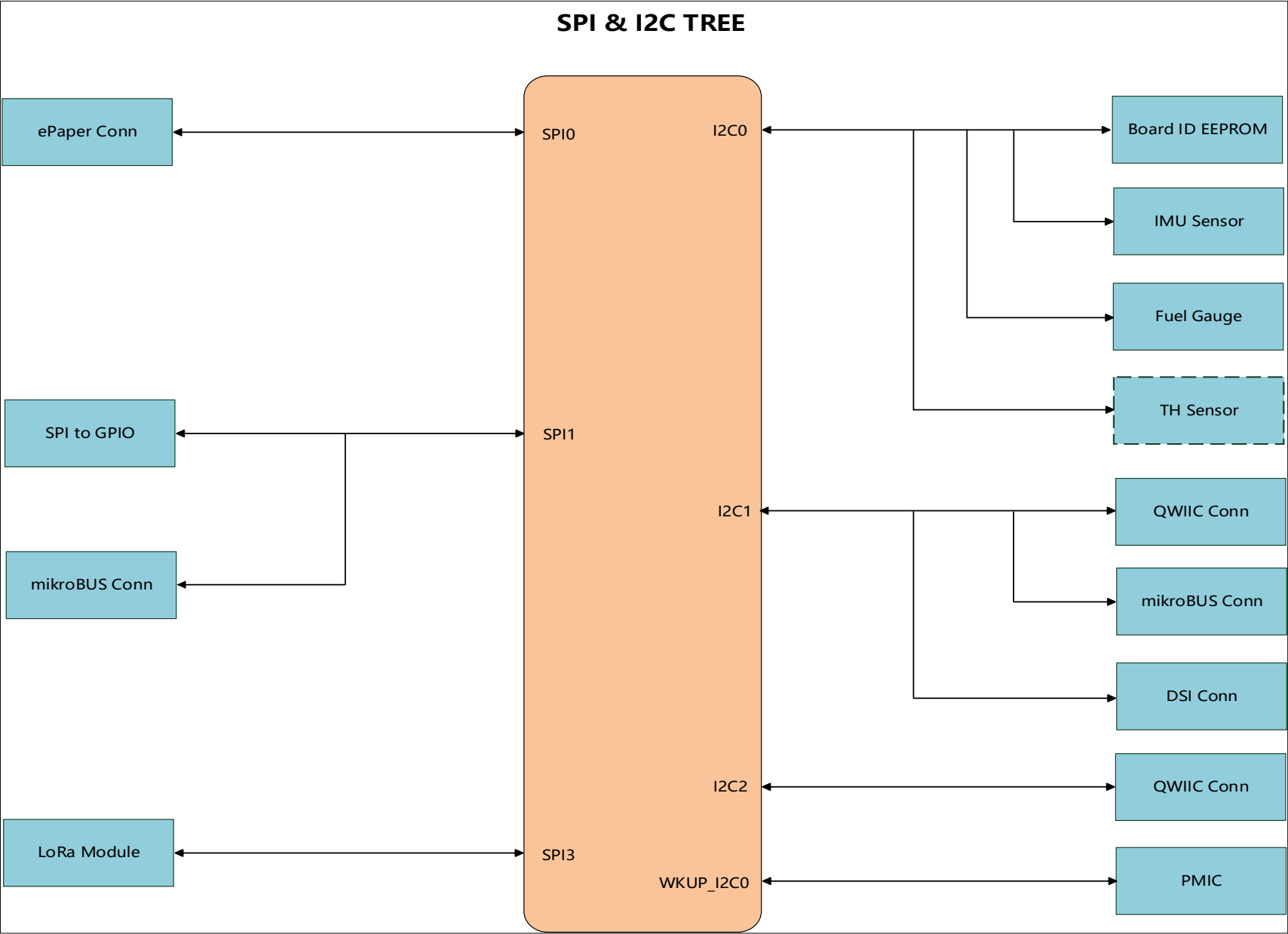
SHEET	SHEET NAME
01	Table of Contents
02	Block Diagram
03	Power Tree
04	IIC Tree
05	Power Input 5V
06	RTC ONLY Mode
07	AM62L PMIC
08	AM62L Power
09	AM62L DCAPs
10	AM62L WKUP & OSC & Reset
11	AM62L BOOT MODE
12	AM62L LPDDR4 DEVICE
13	AM62L OSPI Flash
14	AM62L Micro SD
15	AM62L ADC & DSI & OTHER
16	AM62L USB & HUB
17	USB2.0 Type A Host & UART
18	BM3301 & Sx1262 Module
19	EEPROM & SOICbite
20	Button & LEDs
21	Buzzer & IMU & Light
22	QWIIC & MikroBus
23	ePaper Conn
24	Other PINs & DSI & Grove
25	AM62L eMMC

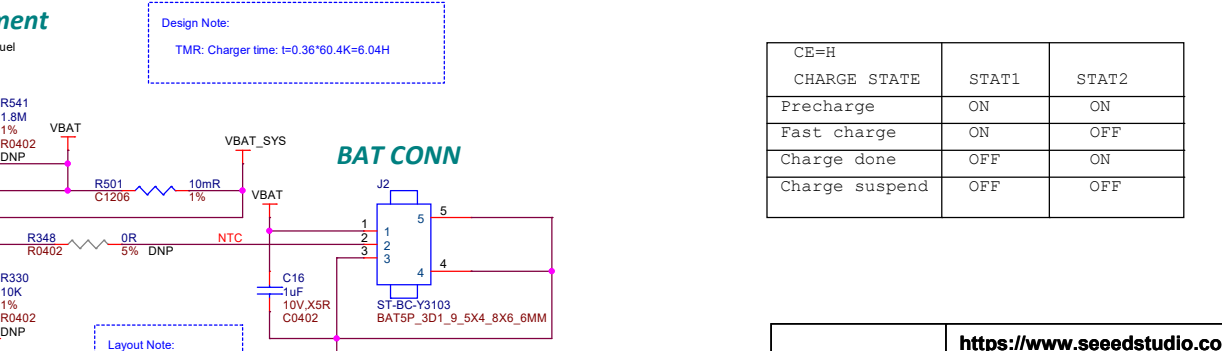
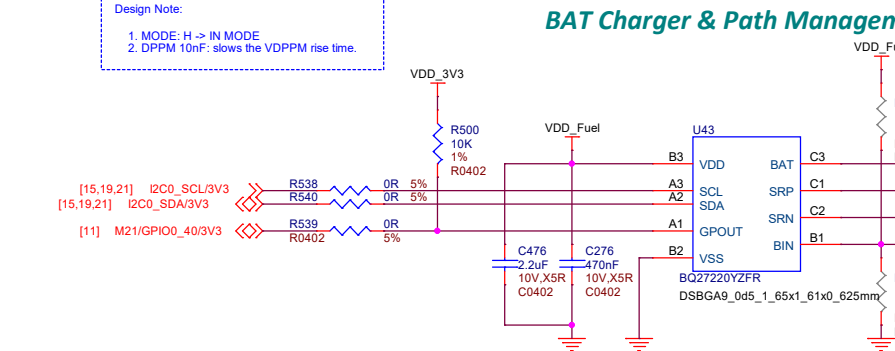
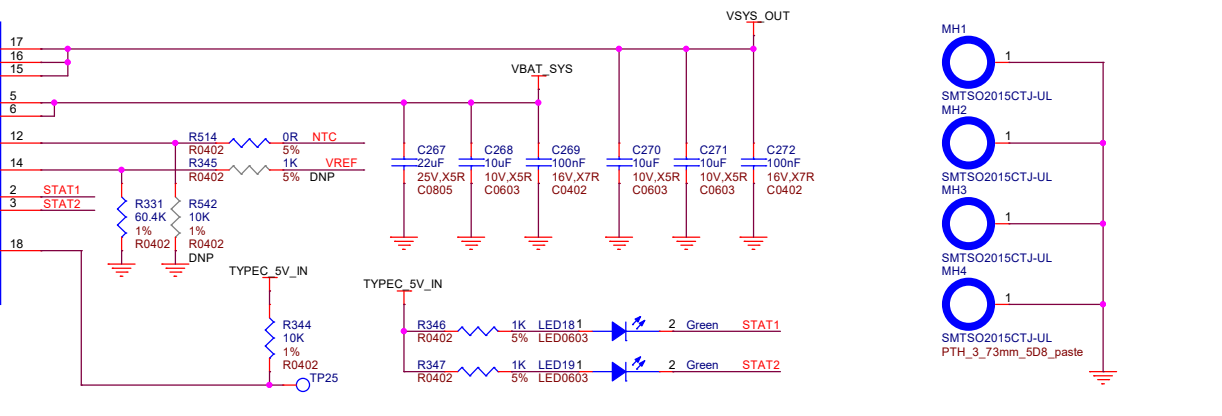
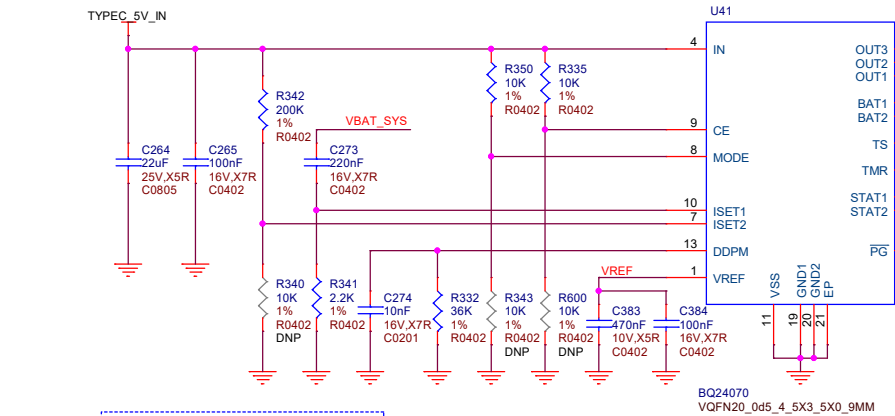
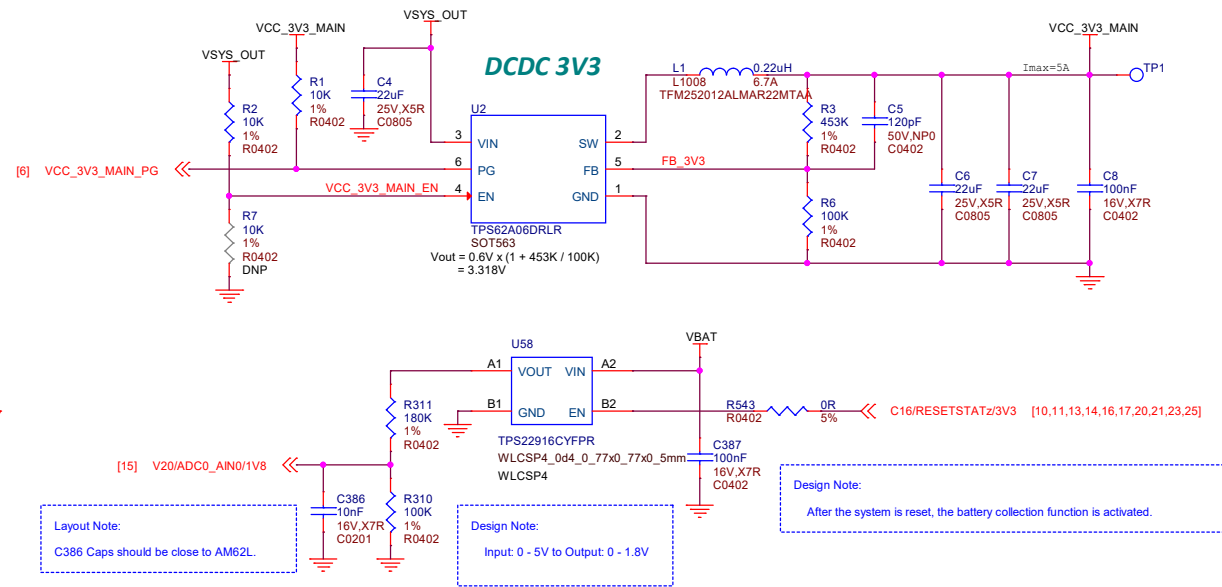
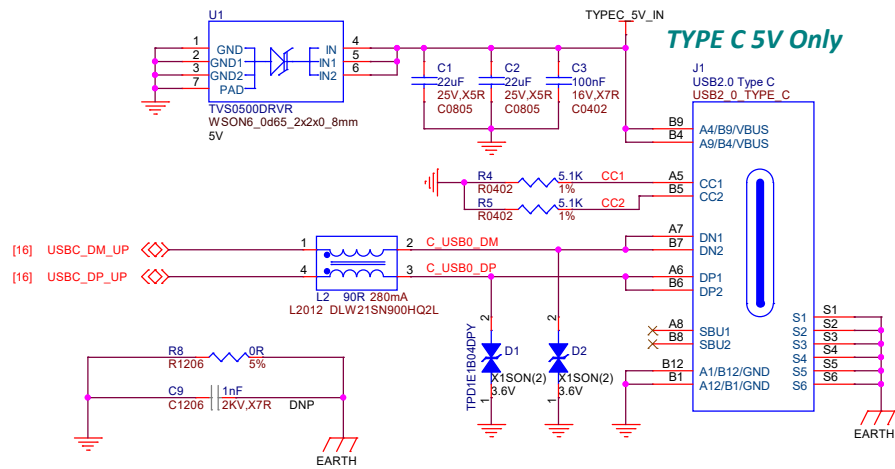
Revision History

DATE	REVISION	DESCRIPTION
2025/10/14	Rev A	1. Initial
2025/10/20	V0.3	Main modifications: 1. Change the capacitor withstand voltage to 2x or higher. 2. Correct the schematic symbol of the AND gate. 3. Add switch control to the power monitoring ADC. 4. Modify the circuit of RTC_PG and PMIC to reset AM62L. 5. DNI the output capacitor of TPS22965DSGR. 6. Add eMMC circuit. 7. Correct the wake-up signal(WAKEUP0/1) . 8. Change Tag-Connect to SOIC16 Bite. 9. Change button gpios to 3V3 IOs because connect to SDIOBite 10. DNI I2C singles unpopulated for compatibility touchscreens.
2025/10/24	V0.4	1. Replace U9 with 0R resistors to switch Low Power mode. 2. Boot and Select function share one button. 3. Replace the Up/Down/Left/Right buttons with a joystick 4. Use the SPI1 to connect U42 (SPI to GPIO) and mikroBUS 5. Correct and add network identification descriptions 6. Adjust some GPIO allocation.
2025/10/28	V0.5	1. Added power supply selection for the IMU. 2. Modified the U62 circuit to prevent its input from floating. 3. Added the EXT_WAKEUP0 signal to pin 12 of J29 (SOICbite) .
2025/10/29	V0.6	1. Use M23/GPIO0_43/3V3 to reset the USB Hub. 2. Use F22/GPIO0_29/PWM/3V3 to drive the buzzer. 3. The RGB LED has been changed to a smaller package.
2025/11/05	V0.7	1. Page8: Correct the VDDS0 power supply voltage to 1V8. 2. Page19: The JTAG signals change to 3.3V through a buffer. 3. Page24: The "Other GPIOs" pins shall be changed to 1V8.

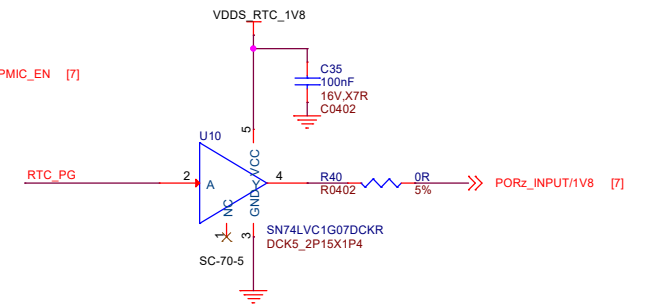
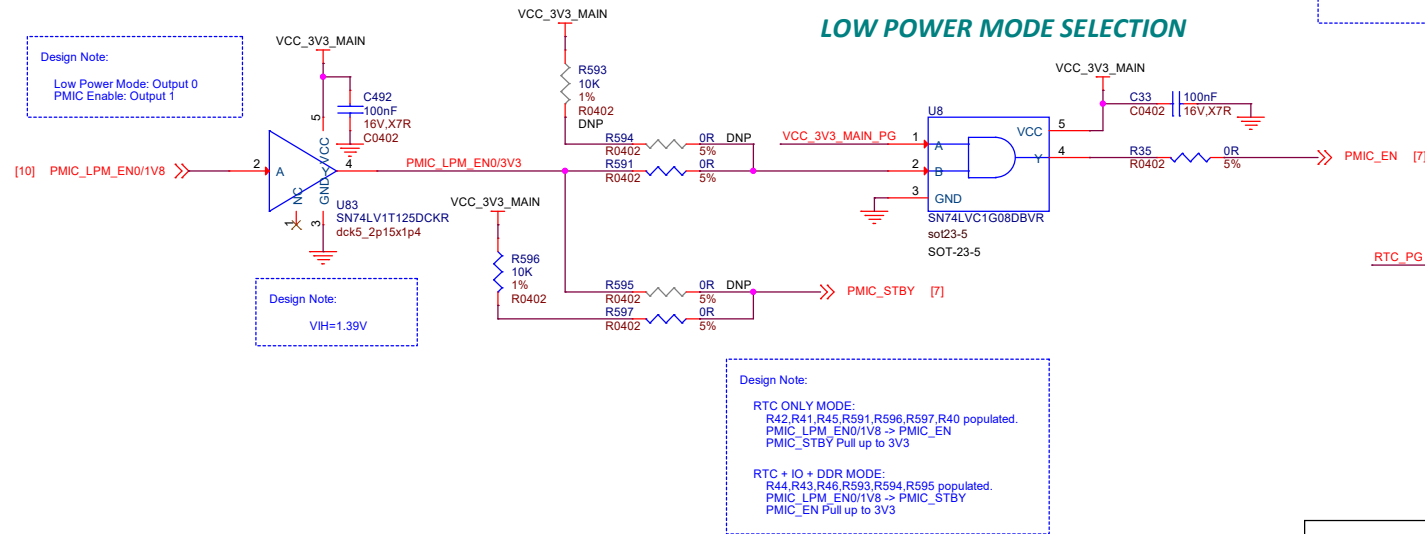
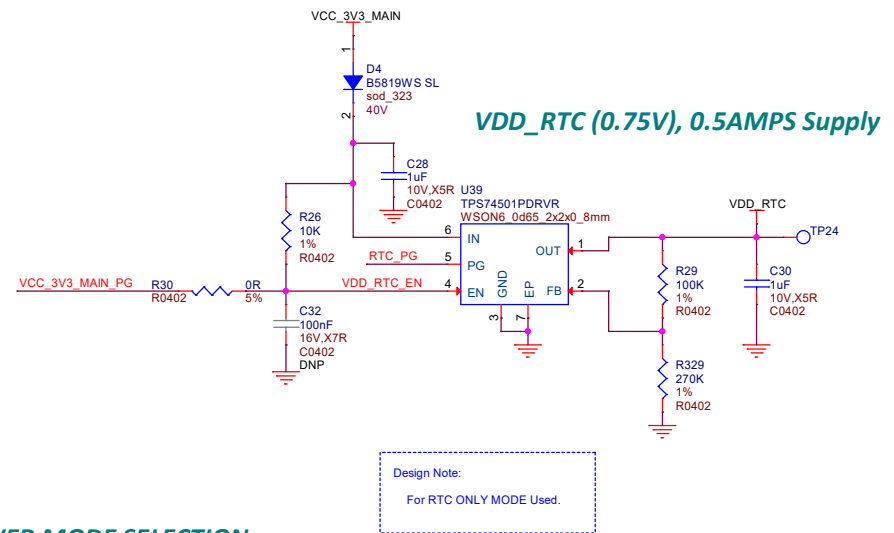
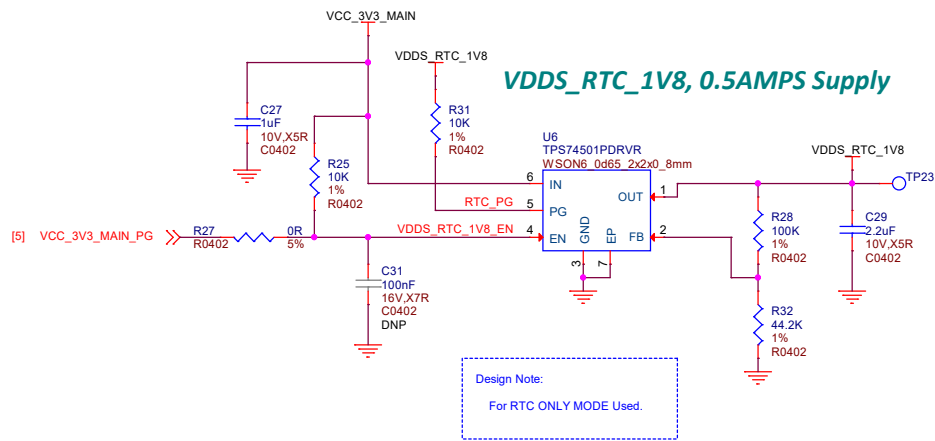




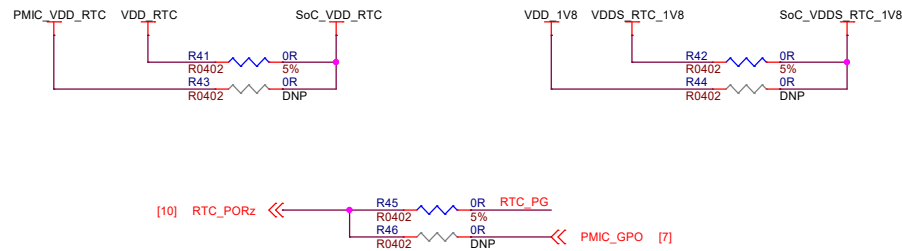




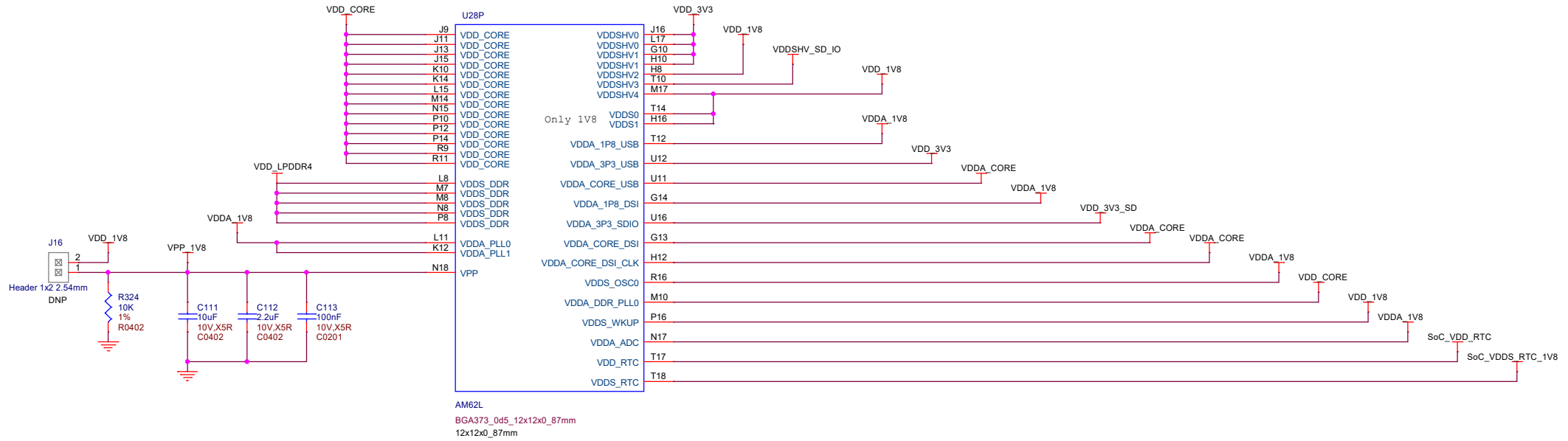
CE=H	STAT1	STAT2
CHARGE STATE	ON	ON
Precharge	ON	ON
Fast charge	ON	OFF
Charge done	OFF	ON
Charge suspend	OFF	OFF



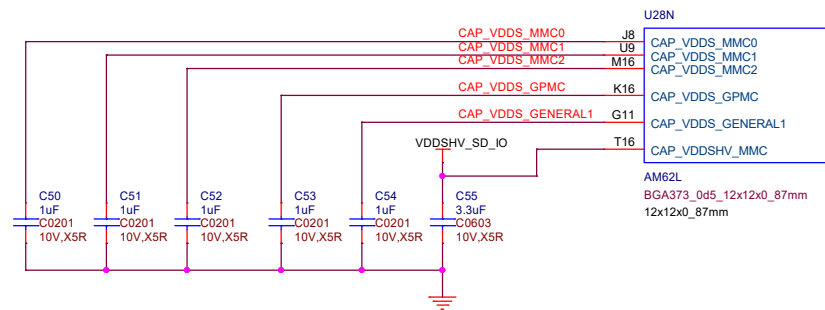
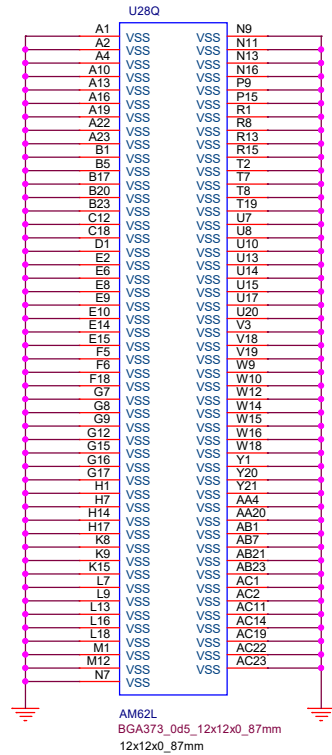
RTC Only MODE	TO SOC	RTC + IO + DDR MODE
VDDS_RTC_1V8	SoC_VDDS_RTC_1V8	SOC_DVDD1V8
VDD_RTC	SoC_VDD_RTC	PMIC_VDD_RTC
RTC_PG	RTC_PORz	PMIC_GPO



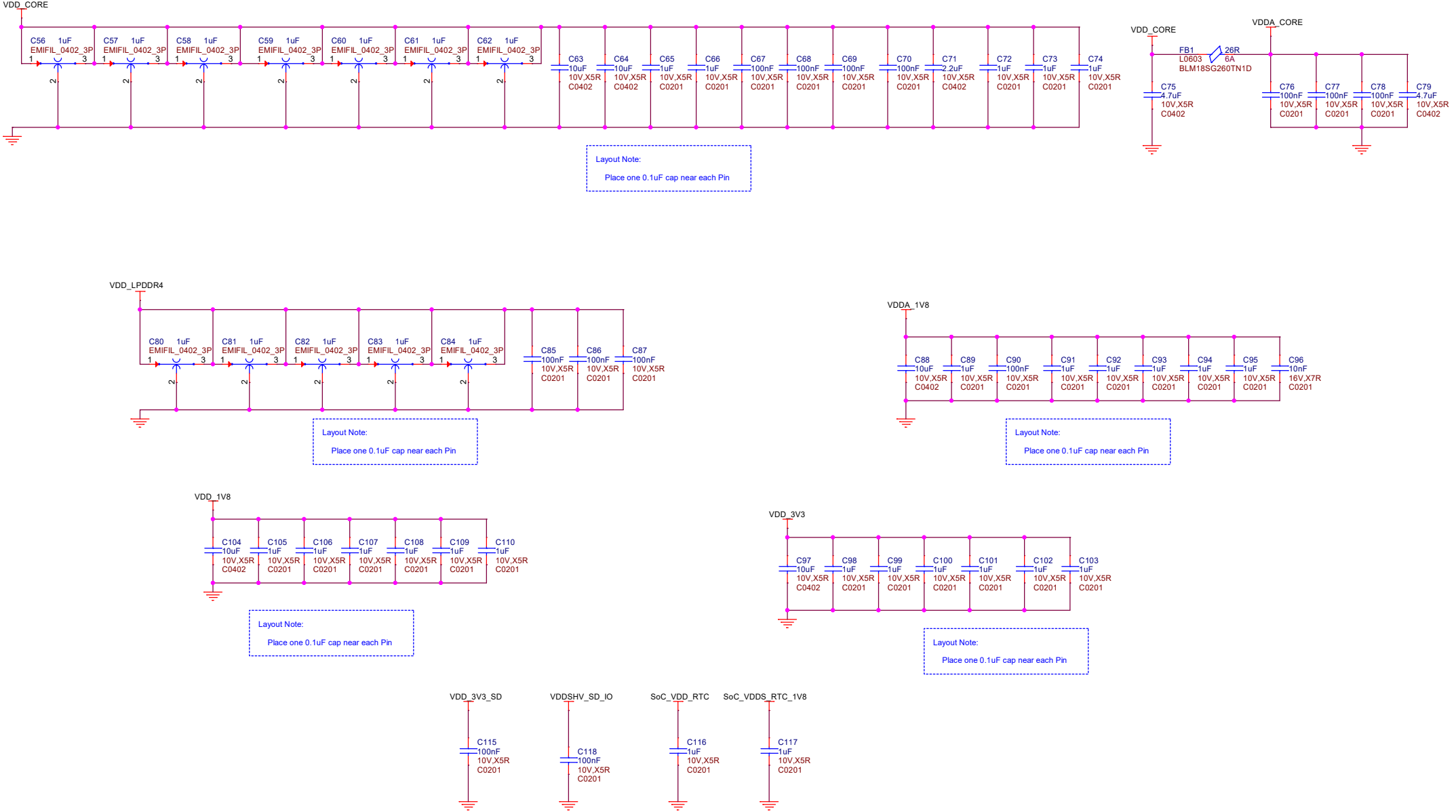
SOC POWER SUPPLIES AND SUPPLY RAILS



SOC VSS



SOC POWER SUPPLIES - DECAPS



SOC WKUP DOMAIN

U28M

VDDSD0
SET: 1V8

MDIO0_MDC
MDIO0_MDIO

WKUP_I2C0_SCL
WKUP_I2C0_SDA

VDDSD_WKUP
SET: 1V8

WKUP_CLKOUT0
WKUP_OSC0_XI

WKUP_OSC0_XO

WKUP_UART0_RTSM
WKUP_UART0_CTSN

WKUP_UART0_RXD
WKUP_UART0_TXD

AM62L

BGA373_0d5_12x12x0_87mm
12x12x0_87mm

AC15 AC15/MDIO0_MDC/1V8 [24]

AC13 AC13/MDIO0_MDIO/1V8 [24]

AB22 AA22

Y23 Y23/CLKOUT_32K/1V8 [18]

AC18 WKUP_OSC0_XI

AC17 WKUP_OSC0_XO

W22 W22/WKUP_GPIO0_3/1V8 [24]

W23 W23/WKUP_GPIO0_2/INT/1V8 [18]

Y22 Y22/WKUP_GPIO0_0/1V8 [24]

AA23 AA23/WKUP_GPIO0_1/1V8 [21]

VDD_1V8

R55 4.7K 5% R0402

R56 4.7K 5% R0402

WKUP_I2C0_SCL [7]

WKUP_I2C0_SDA [7]

R57 R0201 0R 1%

C119 C0201 15pF 50V.NP0

X1

25MHz 10pF

x4_smd_2_0x1_6mm

10ppm,10pF

R338 R0201 0R 1%

C120 C0201 15pF 50V.NP0

R60 0R 1% R0201 DNP

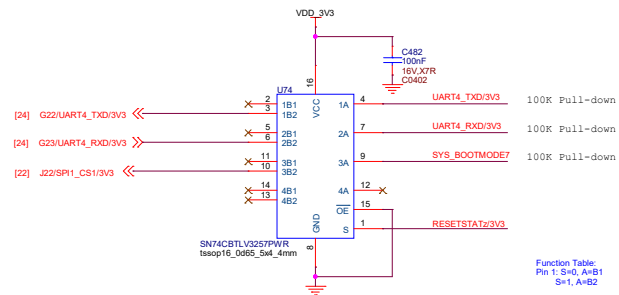
U28C

VDD5_OSC0 PORZ
SET: 1V8
RESETSTATZ
VDD5HV1 RESETZ
SET: 3V3
EXTINTN

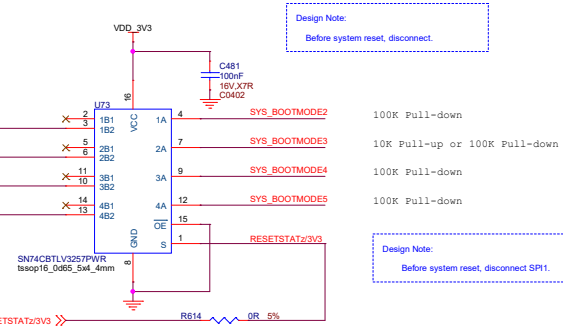
AB18
C16
E16
C8 EXTINTn_RC
R66 R0402
49.9R 1%
C124 100pF
C0201 50V.NPO
C389 22pF
C0402 50V.NPO
R67 10K 1% R0402

PMIC_RSTOUTn/1V8 [7]
C16/RESETSTATz/3V3 [5,11,13,14,16,17,20,21,23,25]
RESETz [19,20]
PMIC_nINT [7]

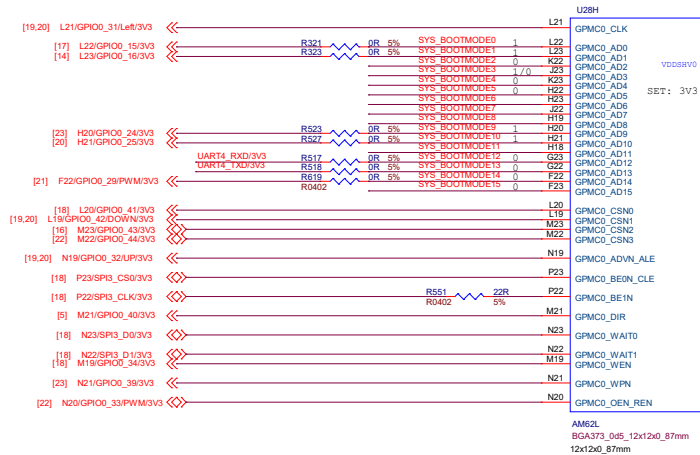
AM62L
BGA373_0d5_12x12x0_87mm
12x12x0_87mm



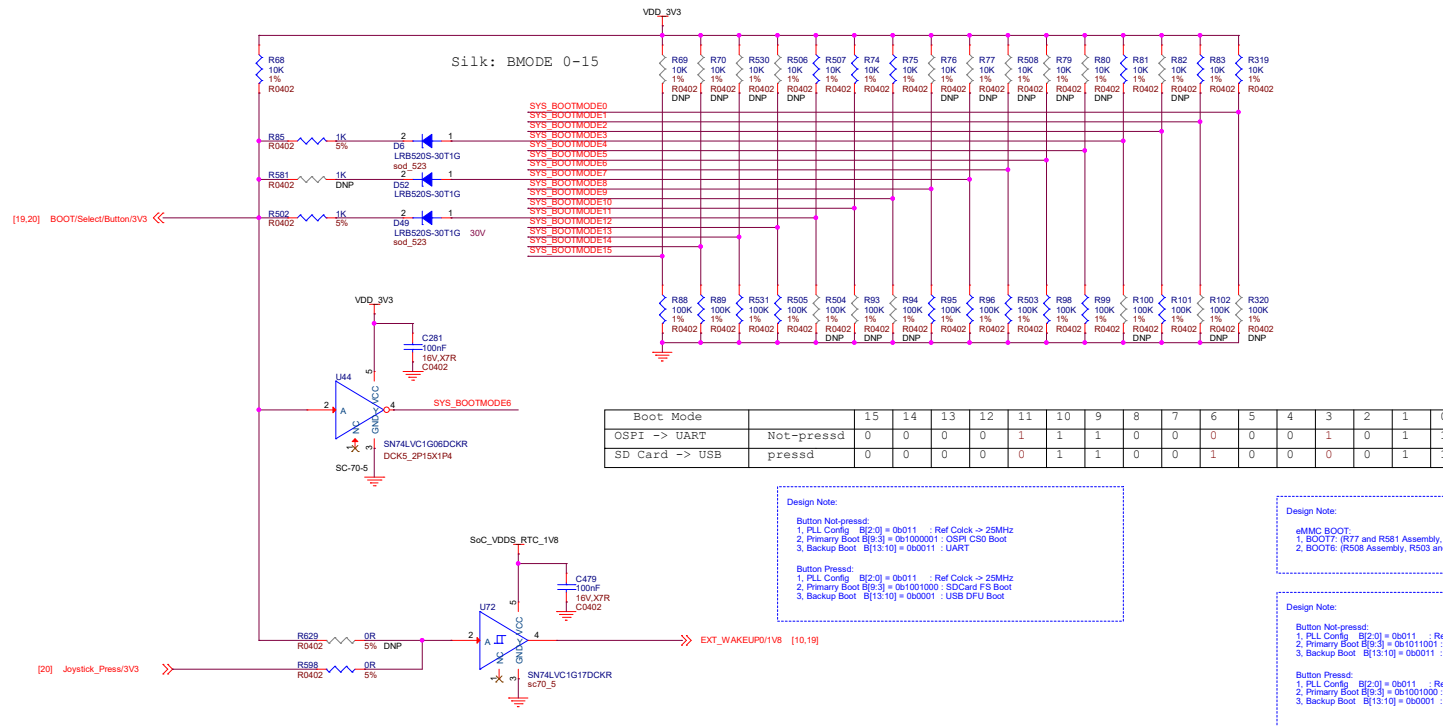
Function Table:
Pin 1: S=0, A=B1
S=1, A=B2



Design Note:
Before system reset, disconnect SPI1.



SOC - GPMC



Boot Mode		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OSPI -> UART	Not-pressd	0	0	0	0	1	1	1	0	0	0	0	0	1	0	1	1
SD Card -> USB	pressd	0	0	0	0	0	1	1	0	0	1	0	0	0	0	1	1

Design Note:
Button Not-pressed:
1. PLL Config [B(2:0)] = 0b011 : Ref Clock -> 25MHz
2. Primary Boot [B(9:3)] = 0b1000001 : OSPI CS0 Boot
3. Backup Boot [B(13:10)] = 0b0011 : UART

Design Note:
eMMC BOOT:
1. BOOT7: (R77 and R581 Assembly, R56 to DNP)
2. BOOT6: (R558 Assembly, R503 and U44 to DNP)

Design Note:
Button Not-pressed:
1. PLL Config [B(2:0)] = 0b011 : Ref Clock -> 25MHz
2. Primary Boot [B(9:3)] = 0b1011001 : eMMC Boot
3. Backup Boot [B(13:10)] = 0b0011 : UART

Boot Mode		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
eMMC -> UART	Not-pressd	0	0	0	0	1	1	1	0	1	1	0	0	1	0	1	1
SD Card -> USB	pressd	0	0	0	0	0	1	1	0	0	1	0	0	0	0	1	1

Design Note:
Use the Joystick_Press Button or Select Button to wake-up System.

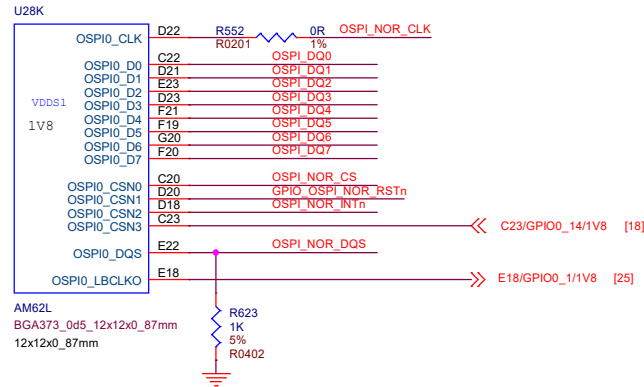
Silk: LPDDR4



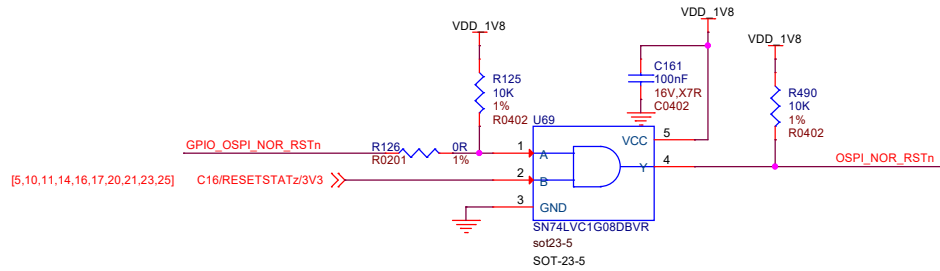
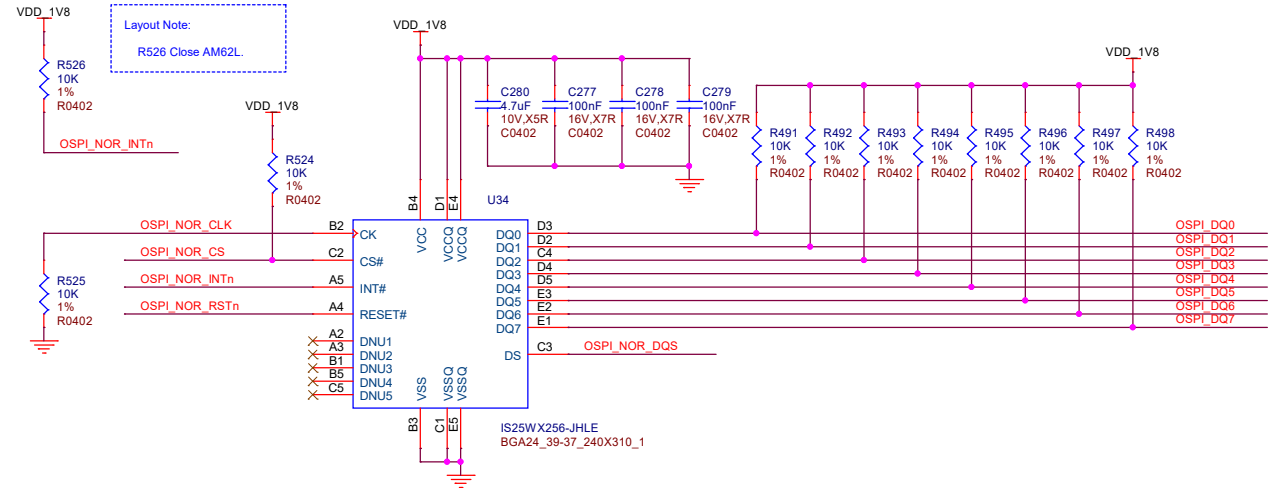
U281
L5 G1 LPDDR4 DQS0 P



SOC OSPI INTERFACE



OSPI NOR FLASH



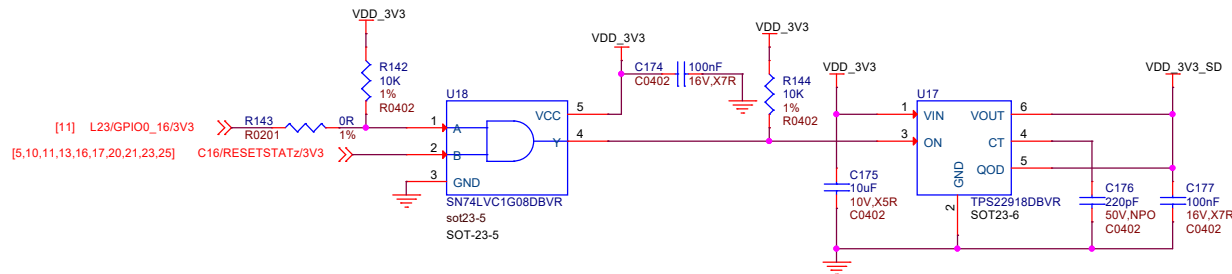
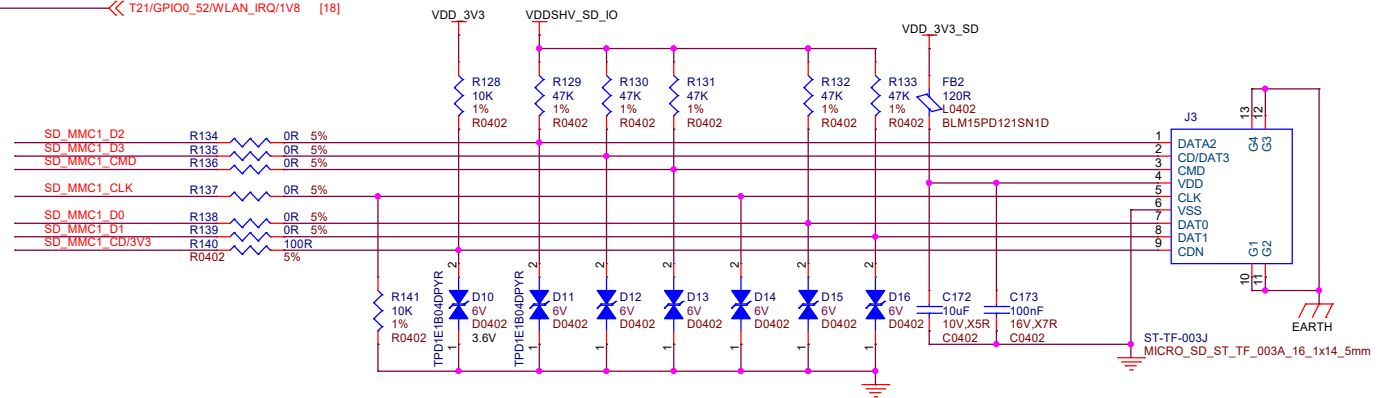
SOC - MMC Interface

U28J



AM62L
BGA373_0d5_12x12x0_87mm
12x12x0_87mm

MICRO SD CARD



U28E



AM62L

BGA373_0d5_12x12x0_87mm
12x12x0_87mm

SOC - MCASP and UART

U28G

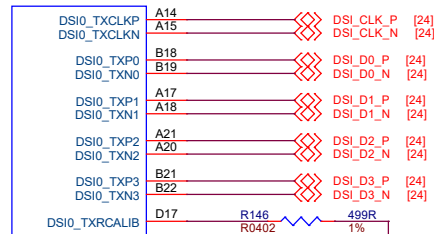


AM62L

BGA373_0d5_12x12x0_87mm
12x12x0_87mm

SOC - DSI

U28O

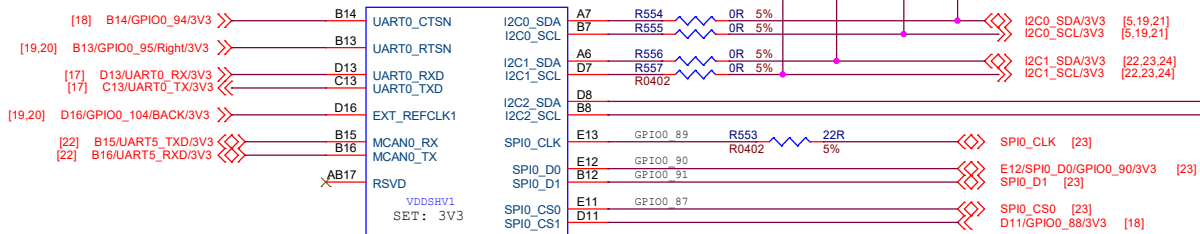


AM62L

BGA373_0d5_12x12x0_87mm
12x12x0_87mm

SOC - UART, MCAN, I2C and IOs

U28A

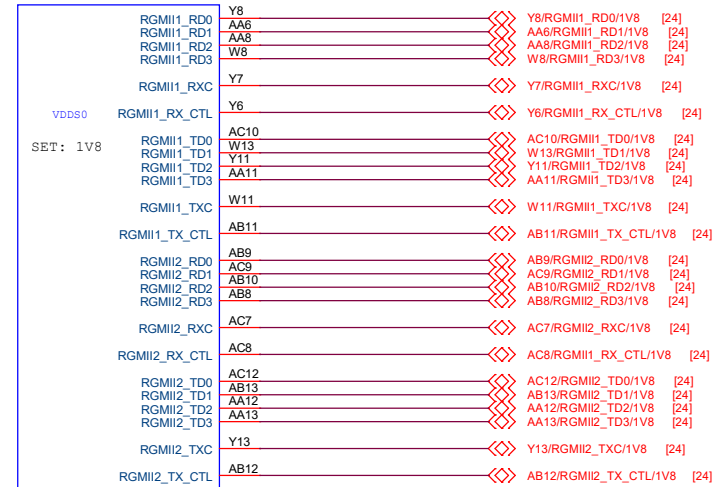


AM62L

BGA373_0d5_12x12x0_87mm
12x12x0_87mm

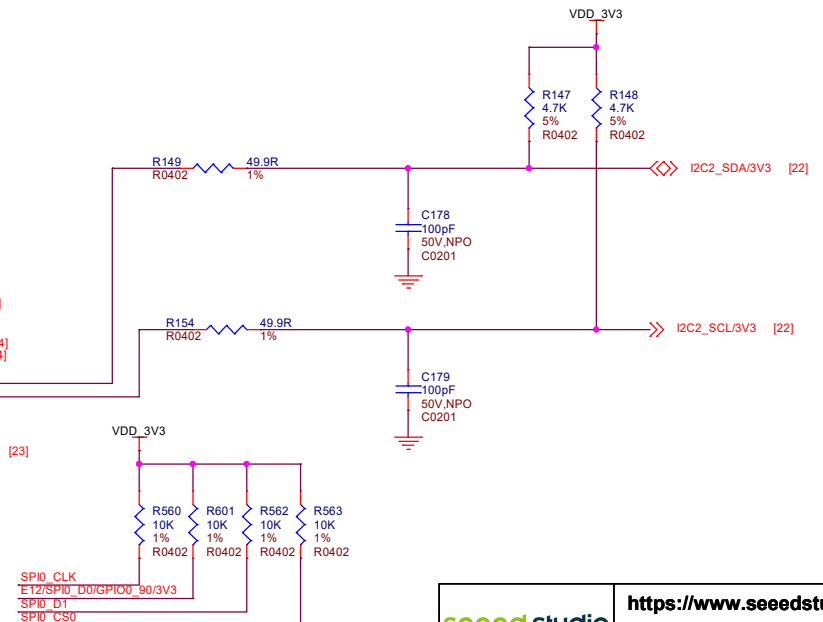
SOC - ETHERNET INTERFACE

U28L



AM62L

BGA373_0d5_12x12x0_87mm
12x12x0_87mm



seeed studio

<https://www.seeedstudio.com>

Title: BeagleBadge AM62L

Size:
A3

Document Number:

15 AM62L ADC & DSI & OTHER

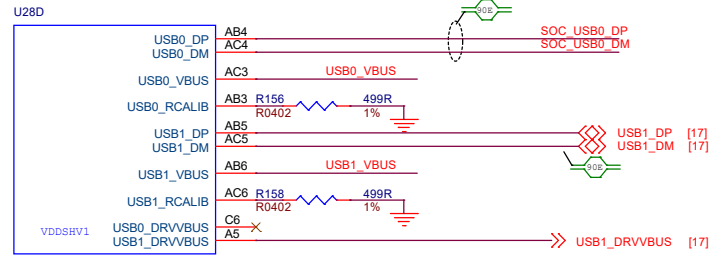
Rev:
RevA

Date:

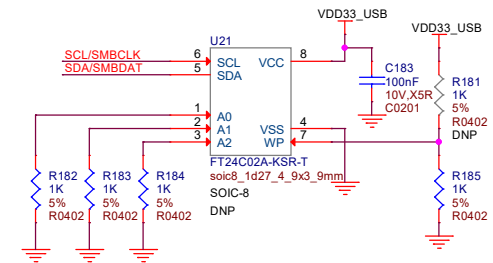
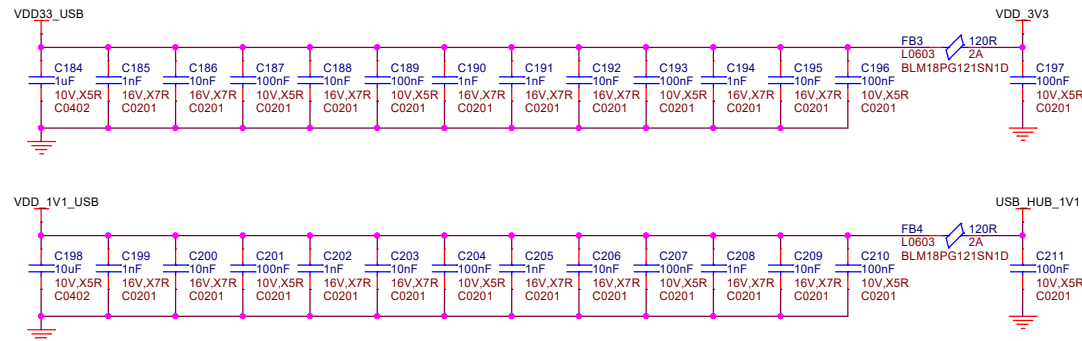
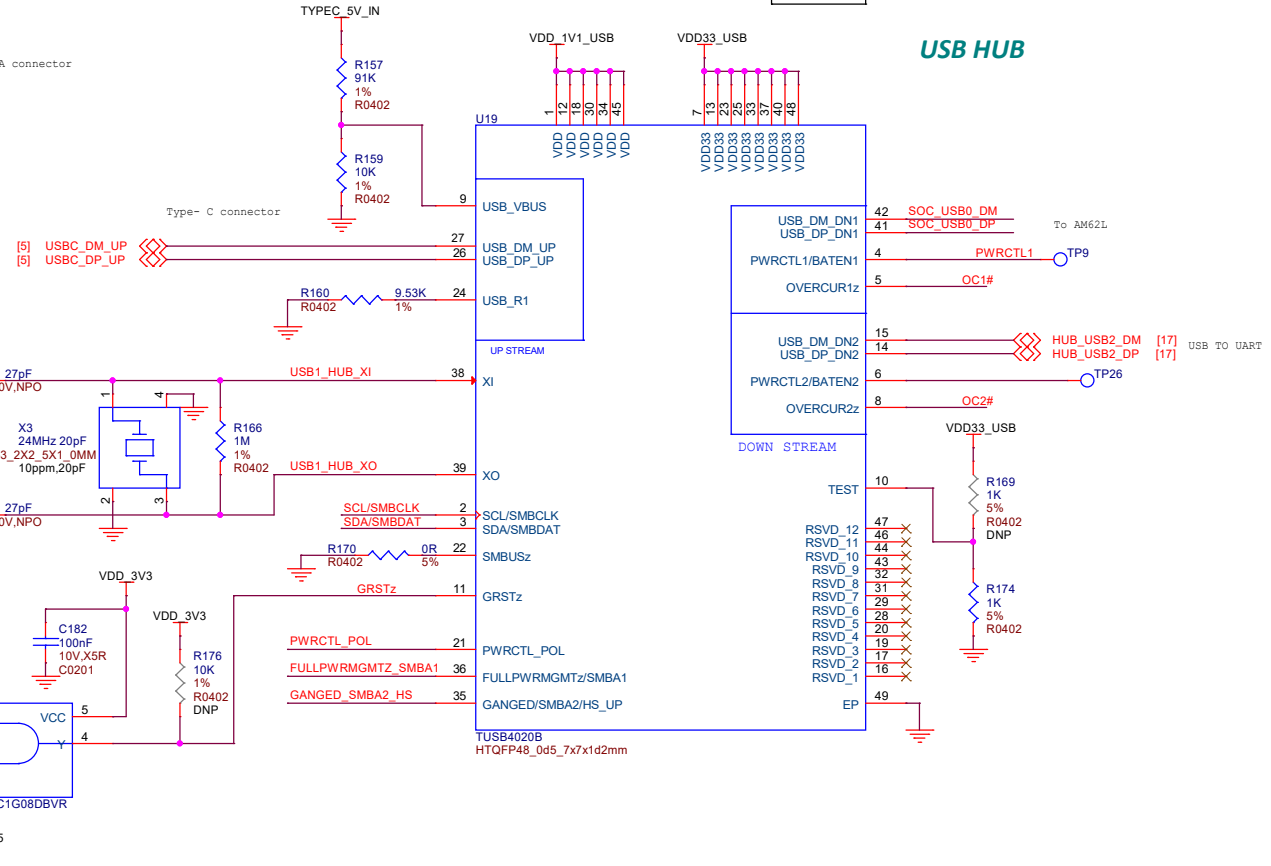
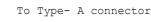
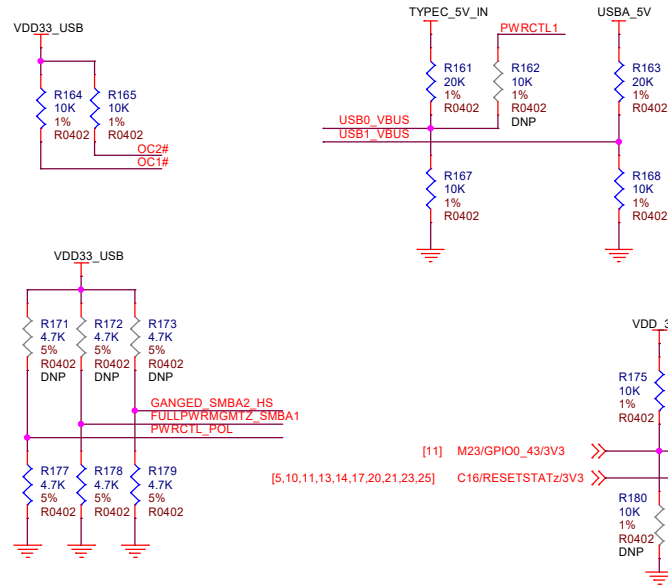
Friday, November 07, 2025

Sheet: 15 of 25

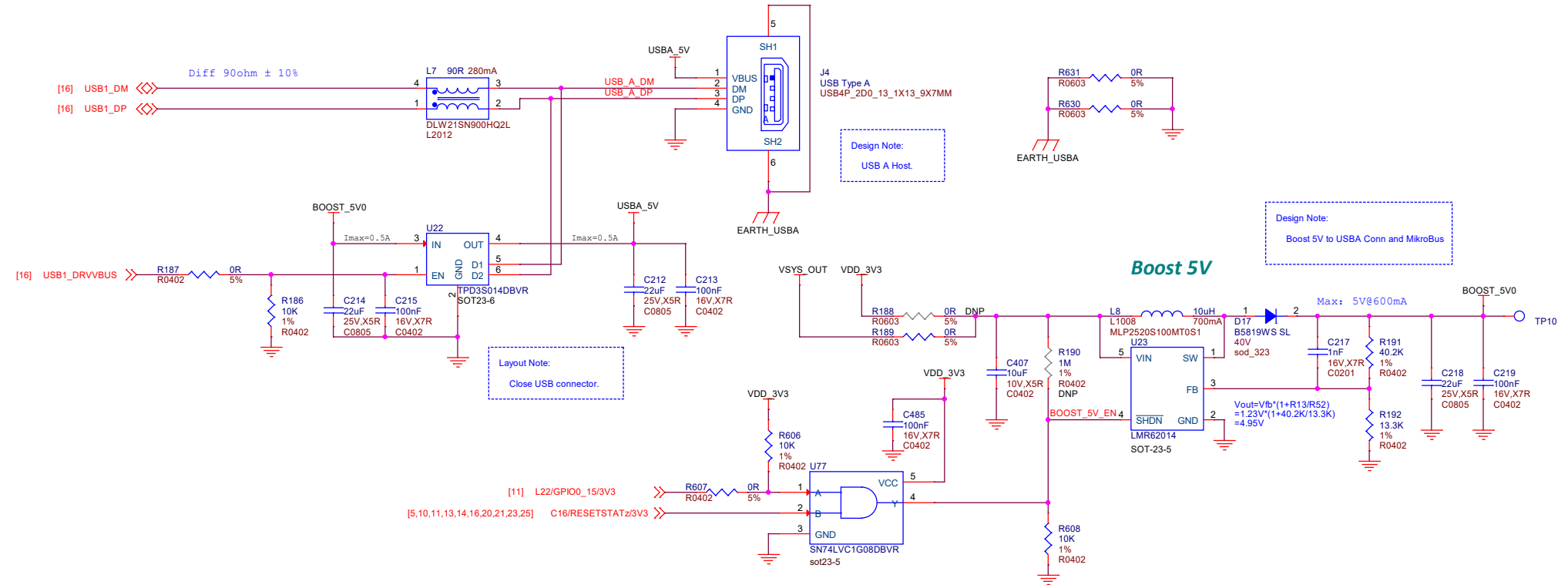
SOC - USB



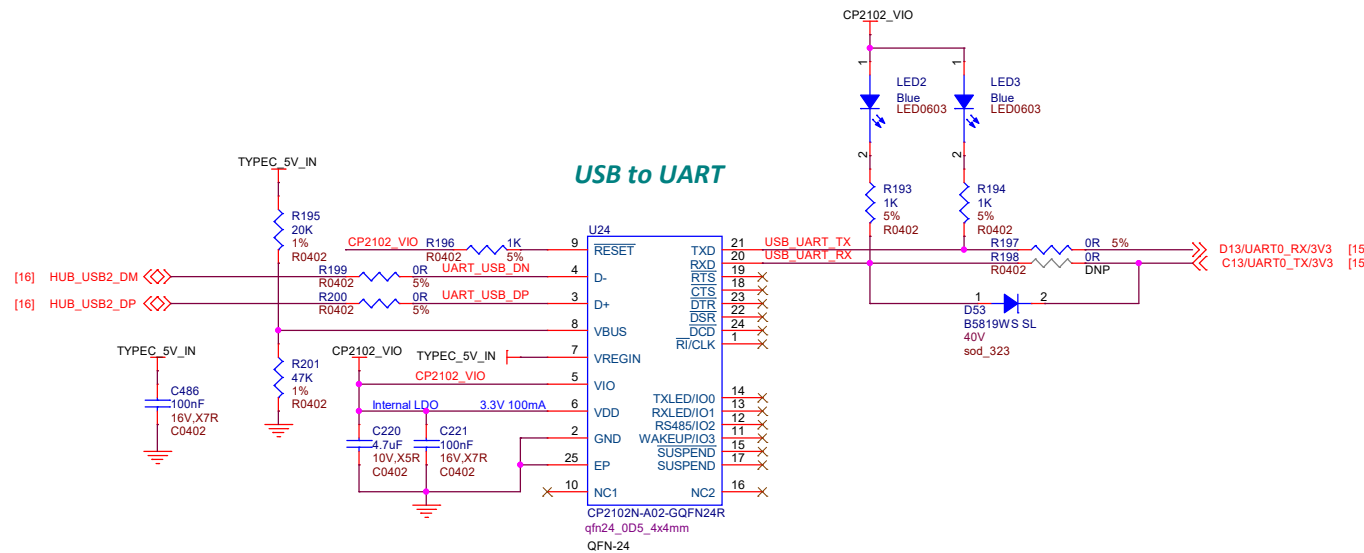
AM62L
BGA373_0d5_12x12x0_87mm
12x12x0_87mm



USB TYPEA



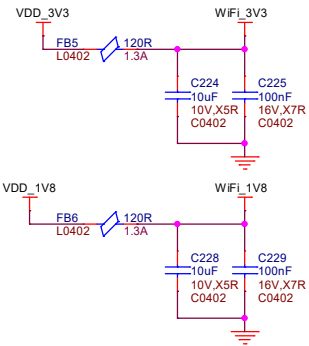
USB to UART



CC3301 Module

Design Note:
AM62L SDIO Signals to WIFI.

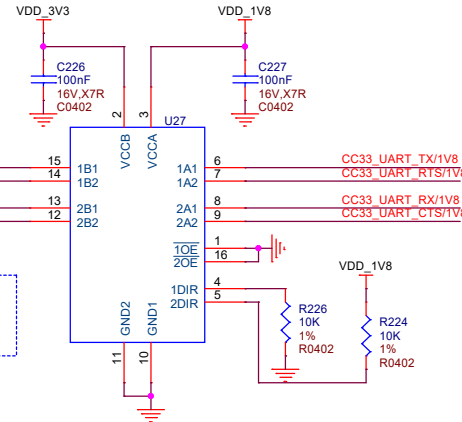
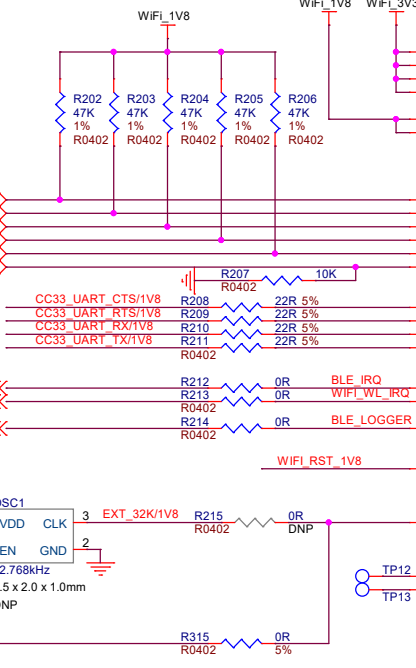
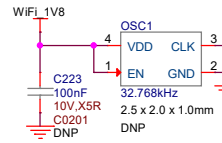
Design Note:
AM62L UART Signals to BLE.



- [14] WIFI_SDIO_D0/1V8
- [14] WIFI_SDIO_D1/1V8
- [14] WIFI_SDIO_D2/1V8
- [14] WIFI_SDIO_D3/1V8
- [14] WIFI_SDIO_CMD/1V8
- [14] WIFI_SDIO_CLK/1V8

- [10] W23/WKUP_GPIO0_2/INT/1V8
- [14] T21/GPIO0_52/WLAN_IRQ/1V8
- [13] C23/GPIO0_14/1V8

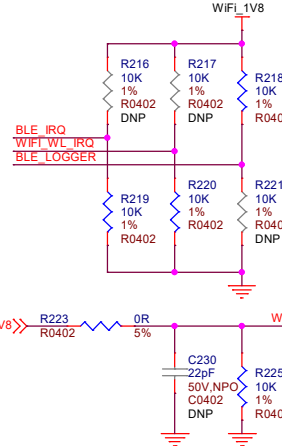
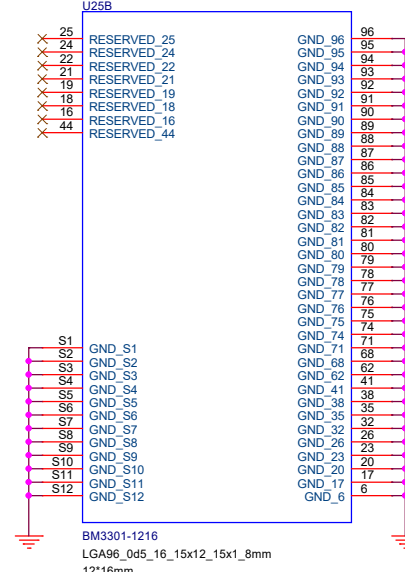
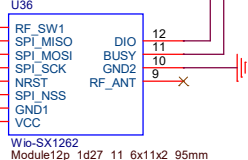
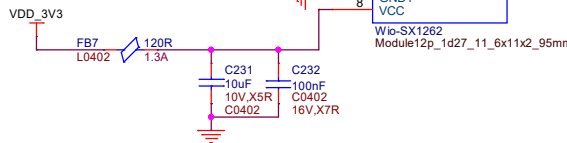
- [10] Y23/CLKOUT_32K/1V8



Design Note:
DIR = 0 : B to A
DIR = 1 : A to B

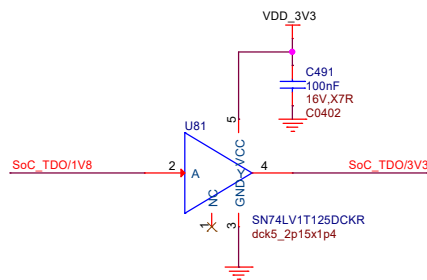
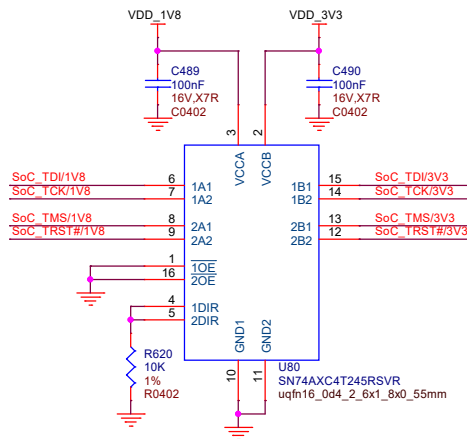
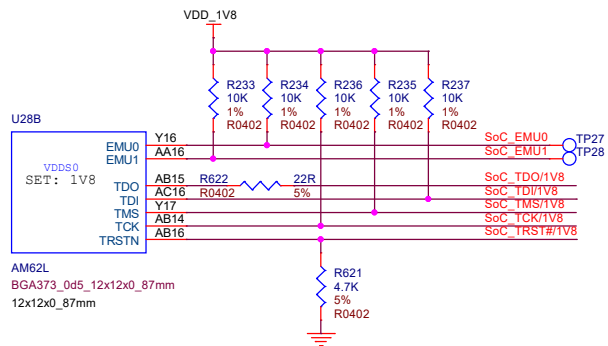
LoRa Module

- [15] D11/GPIO0_88/3V3
- [15] B14/GPIO0_94/3V3
- [11] L20/GPIO0_41/3V3
- [11] N23/SP13_D0/3V3
- [11] N22/SP13_D1/3V3
- [11] P22/SP13_CLK/3V3
- [11] M19/GPIO0_34/3V3
- [11] P23/SP13_CS0/3V3

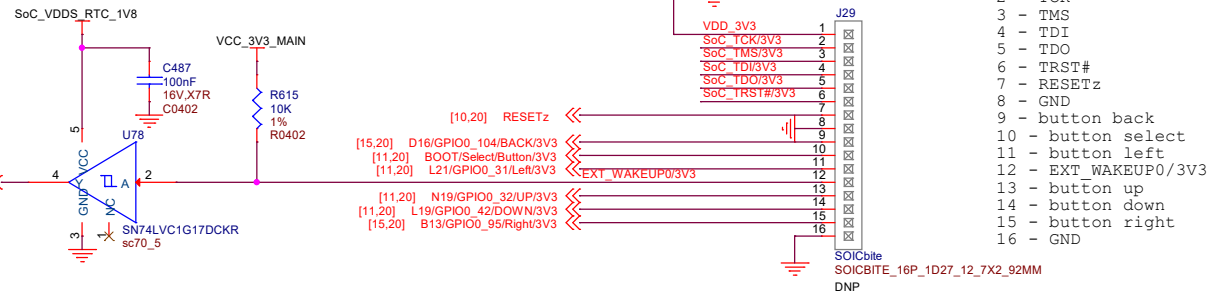


Design Note:
DEFAULT: WIFI and BLE Close.

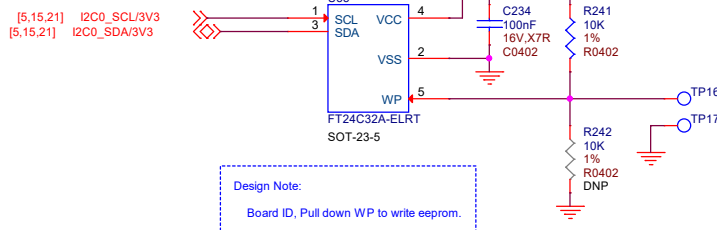
SOC-JTAG



SOICbrite



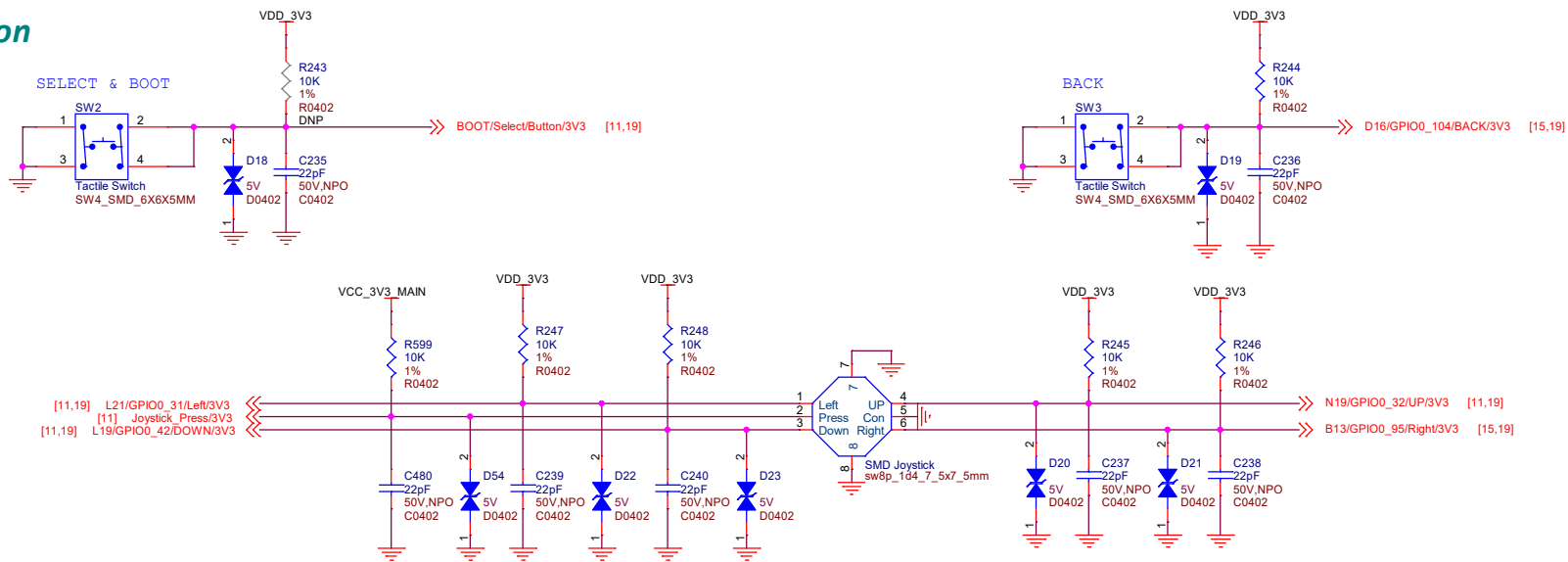
- Pins
- 1 - VDD
 - 2 - TCK
 - 3 - TMS
 - 4 - TDI
 - 5 - TDO
 - 6 - TRST#
 - 7 - RESETz
 - 8 - GND
 - 9 - button back
 - 10 - button select
 - 11 - button left
 - 12 - EXT_WAKEUP0/3V3
 - 13 - button up
 - 14 - button down
 - 15 - button right
 - 16 - GND



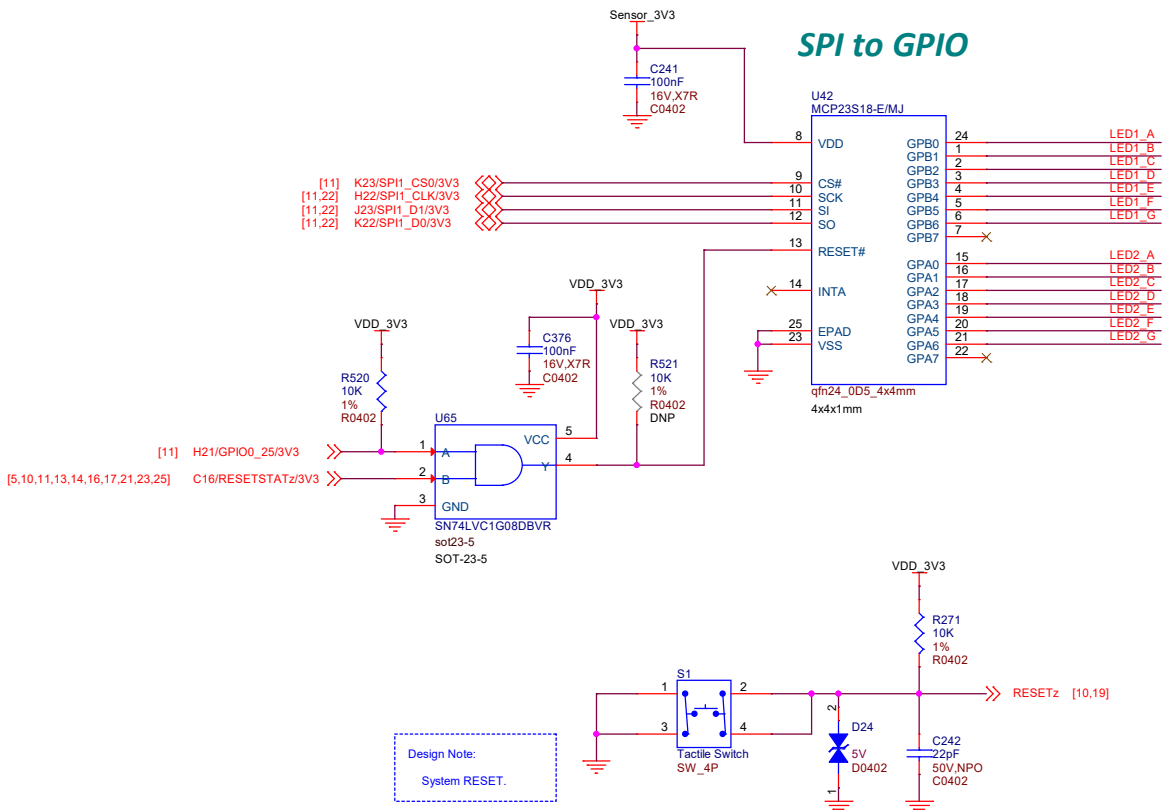
Design Note:
Board ID, Pull down WP to write eeprom.

BOARD ID EEPROM 32Kbit

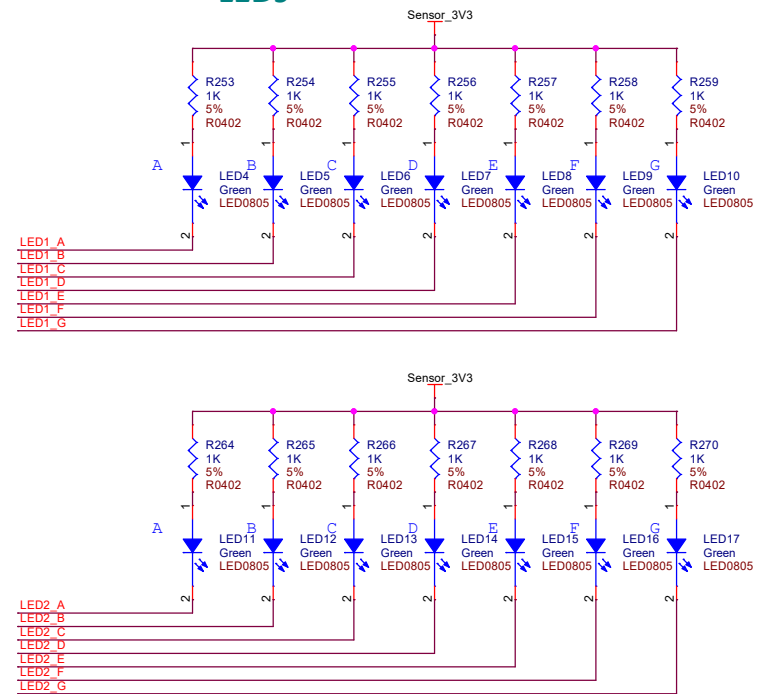
Button



SPI to GPIO



LEDs



Sensor Power Switch

VDD_1V8

R272 10K 1% R0402

U61 VCC

SN74LVC1G08DBVR sot23-5

VDD_1V8

C243 100nF 16V.X7R C0402

VDD_3V3

R273 10K 1% R0402

C244 10uF 10V.X5R C0402

U32 VIN VOUT CT QD ON

TPS22918DBVR SOT23-6

Sensor_3V3

C245 220pF 50V.NPO C0402

C406 10uF 10V.X5R C0402

C391 100nF 16V.X7R C0402

Design Note:

V_{IH} min = 1V.

RGB LED

Sensor_3V3

D25 RGB LED_4P_1_6X1_6MM_RGB

R276 1K 5% R0402

R277 1K 5% R0402

R278 2.2K 1% R0402

Q1 2SK3018 SOT-323-3

Q2 2SK3018 SOT-323-3

Q3 2SK3018 SOT-323-3

R282 100K 1% R0402

R283 100K 1% R0402

R284 100K 1% R0402

[15] A11/GPIO0_83/PWM/3V3

[15] A9/GPIO0_81/PWM/3V3

[15] B9/GPIO0_82/PWM/3V3

Passive Buzzer

VCC_3V3_MAIN VDD_3V3

R616 0R 5% DNP

R617 0R 5% DNP

VCC_3V3_IMU

[5,15,19,21] I2C0_SCL/3V3

[5,15,19,21] I2C0_SDA/3V3

R287 0R 5% IMU_SCL

R289 0R 5% IMU_SDA

VCC_3V3_MAIN

R290 0R 5% DNP

R291 0R 5% DNP

R292 0R 5% DNP

R294 0R 5% DNP

VCC_3V3_IMU

R295 100K 1% R0402

[22] IMU_INT/3V3

Software: Default: High INT: low

VCC_3V3_IMU

R288 4.7K 5% R0402

U33 SDO/SA0 SCL SDA SDx SCx INT1 VDDIO VDD GND1 GND2

LSM6DS3TR-C LGA14_0d5_2_5x3x0_86mm

VCC_3V3_IMU

C248 100nF 16V.X7R C0402

IMU Sensor

T & H Sensor

[5,15,19,21] I2C0_SDA/3V3

[5,15,19,21] I2C0_SCL/3V3

R326 0R 5% R0402

R325 0R 5% R0402

U40 SDA VDD SCL VSS GND_EP

SHT40 dfn4_0d8_1_5x1_5x0_54mm_rko DNP

Sensor_3V3

R327 0R 5% R0402

C410 100nF 16V.X7R C0402

Light Sensor

Sensor_3V3

C411 100nF 16V.X7R C0402

U45 TLV9061IDPWR X2SON4_0d48_0_8x0_8x0_4mm

R511 10K 1% R0402

R513 12K 1% R0402

C282 10nF 16V.X7R C0201

U45 TLV9061IDPWR X2SON4_0d48_0_8x0_8x0_4mm

R573 0R 5% R0402

Q7 ALS-PT19-315C/L177/TR8

SNR2_1d5_1_7x0_8x0_6mm

R509 100K 1% R0402

Design Note:

V_{out}(max.) = V_{cc} - 0.4V

Layout Note:

C282 Caps should be close to AM62L.

Design Note:

Input: 0 - 2V₉ to Output: 0 - 1V₆

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Title: BeagleBadge AM62L

Size: A3

Document Number: 21 Buzzer & IMU & Light

Date: Friday, November 07, 2025

Rev: RevA

Sheet: 21 of 25

Sensor Power Switch

VDD_1V8

R272 10K 1% R0402

U61 VCC

SN74LVC1G08DBVR sot23-5

VDD_1V8

C243 100nF 16V.X7R C0402

VDD_3V3

R273 10K 1% R0402

C244 10uF 10V.X5R C0402

U32 VIN VOUT CT QD

TPS22918DBVR SOT23-6

Sensor_3V3

C245 220pF 50V.NPO C0402

C406 10uF 10V.X5R C0402

C391 100nF 16V.X7R C0402

Design Note:

V_{IH} min = 1V.

RGB LED

Sensor_3V3

R276 1K 5% R0402

Q1 2SK3018 SOT-323-3

R282 100K 1% R0402

R277 1K 5% R0402

Q2 2SK3018 SOT-323-3

R283 100K 1% R0402

R278 2.2K 1% R0402

Q3 2SK3018 SOT-323-3

R284 100K 1% R0402

D25 RGB LED_4P_1_6X1_6MM_RGB

Passive Buzzer

VCC_3V3_MAIN VDD_3V3

R616 0R DNP

R617 0R DNP

VCC_3V3_IMU

R279 1K 5% R0402

Q4 BC817W SOT323

R281 10K 1% R0402

SMD Buzzer 3.6V V_{op}=2.5-4.5V;4P-8.5*8.5*3.0mm buzzer4p_7d1_8_5x8_5x3mm

IMU Sensor

VCC_3V3_MAIN

R287 0R IMU_SCL

R289 0R IMU_SDA

VCC_3V3_IMU

R290 0R DNP

R291 0R DNP

R292 0R DNP

R294 0R DNP

VCC_3V3_IMU

R295 100K 1% R0402

U33 SDO/SA0 SCL SDA

LSM6DS3TR-C LGA14_0d5_2_5x3x0_86mm

VCC_3V3_IMU

C248 100nF 16V.X7R C0402

C249 100nF 16V.X7R C0402

Software: Default: High INT: low

T & H Sensor

[5,15,19,21] I2C0_SDA/3V3

[5,15,19,21] I2C0_SCL/3V3

R326 0R 5% R0402

R325 0R 5% R0402

U40 SDA VDD SCL VSS GND_EP

SHT40 dfn4_0d8_1_5x1_5x0_54mm_rko DNP

Sensor_3V3

R327 0R 5% R0402

C410 100nF 16V.X7R C0402

Light Sensor

Sensor_3V3

C411 100nF 16V.X7R C0402

U45 TLV9061IDPWR X2SON4_0d48_0_8x0_8x0_4mm

R511 10K 1% R0402

R513 12K 1% R0402

C282 10nF 16V.X7R C0201

U45

R573 0R 5% R0402

Sensor_3V3

Q7 ALS-PT19-315C/L177/TR8

SNR2_1d5_1_7x0_8x0_6mm

R509 100K 1% R0402

Design Note:

V_{out}(max.) = V_{cc} - 0.4V

Layout Note:

C282 Caps should be close to AM62L.

Design Note:

Input: 0 - 2V₉ to Output: 0 - 1V₆

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Title: BeagleBadge AM62L

Size: A3

Document Number: 21 Buzzer & IMU & Light

Date: Friday, November 07, 2025

Rev: RevA

Sheet: 21 of 25

Sensor Power Switch

VDD_1V8, VDD_3V3, Sensor_3V3

[10] AA23/WKUP_GPIO0_1/1V8 >> R274 10K R0402

[5,10,11,13,14,16,17,20,23,25] C16/RESETSTAT2/3V3 >> R272 10K R0402

U61 SN74LVC1G08DBVR SOT-23-5

U32 TPS22918DBVR SOT23-6

C243 100nF C0402

C244 10uF C0402

C245 220pF C0402

C406 10uF C0402

C391 100nF C0402

Design Note: VIH min = 1V.

RGB LED

Sensor_3V3

D25 RGB LED_4P_1_6X1_6MM_RGB

R276 1K R0402

R277 1K R0402

R278 2.2K R0402

Q1 2SK3018 SOT-323-3

Q2 2SK3018 SOT-323-3

Q3 2SK3018 SOT-323-3

R282 100K R0402

R283 100K R0402

R284 100K R0402

[15] A11/GPIO0_83/PWM/3V3 >> R280 1K R0402

[15] A9/GPIO0_81/PWM/3V3 >> R285 1K R0402

[15] B9/GPIO0_82/PWM/3V3 >> R286 1K R0402

Passive Buzzer

VCC_3V3_MAIN VDD_3V3

VCC_3V3_IMU

R616 0R R0402

R617 0R R0402

[11] F22/GPIO0_29/PWM/3V3 >> R279 1K R0402

Q4 BC817W SOT323

R281 10K R0402

SMD Buzzer 3.6V Vop=2.5-4.5V;4P-8.5*8.5*3.0mm buzzer4p_7d1_8_5x8_5x3mm

IMU Sensor

VCC_3V3_MAIN

VCC_3V3_IMU

[5,15,19,21] I2C0_SCL/3V3 >> R287 0R R0402

[5,15,19,21] I2C0_SDA/3V3 >> R289 0R R0402

R290 0R R0402

R291 0R R0402

R292 0R R0402

R294 0R R0402

R295 100K R0402

[22] IMU_INT/3V3 >> R296 0R R0402

U33 LSM6DS3TR-C LGA14_0d5_2_5x3x0_86mm

C248 100nF C0402

C249 100nF C0402

Software: Default: High INT: low

T & H Sensor

[5,15,19,21] I2C0_SDA/3V3 >> R326 0R R0402

[5,15,19,21] I2C0_SCL/3V3 >> R325 0R R0402

U40 SHT40 dfn4_0d8_1_5x1_5x0_54mm_rko DNP

R327 0R R0402

C410 100nF C0402

Light Sensor

Sensor_3V3

C411 100nF C0402

U45 TLV9061IDPWR X2SON4_0d48_0_8x0_8x0_4mm

R511 10K R0402

R513 12K R0402

R550 0R R0402

R509 100K R0402

Q7 ALS-PT19-315C/L177/TR8

SNR2_1d5_1_7x0_8x0_6mm

Design Note: Vout(max.) = Vcc - 0.4V

Layout Note: C282 Caps should be close to AM62L.

Design Note: Input: 0 - 2V9 to Output: 0 - 1V6

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Title: BeagleBadge AM62L

Size: A3

Document Number: 21 Buzzer & IMU & Light

Date: Friday, November 07, 2025

Rev: RevA

Sheet: 21 of 25

Sensor Power Switch

VDD_1V8

VDD_3V3

Sensor_3V3

Design Note: V_{IH} min = 1V.

RGB LED

Sensor_3V3

Design Note: V_{out}(max.) = V_{cc} - 0.4V

Passive Buzzer

VCC_3V3_MAIN VDD_3V3

VCC_3V3_IMU

IMU Sensor

Software: Default: High INT: low

T & H Sensor

Light Sensor

Design Note: C282 Caps should be close to AM62L.

Design Note: Input: 0 - 2V₉ to Output: 0 - 1V₆

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BeagleBadge AM62L

Size: A3 Document Number: 21 Buzzer & IMU & Light Rev: RevA

Date: Friday, November 07, 2025 Sheet: 21 of 25

Sensor Power Switch

VDD_1V8

R272 10K 1% R0402

U61 VCC

SN74LVC1G08DBVR sot23-5

VDD_1V8

C243 100nF 16V.X7R C0402

VDD_3V3

R273 10K 1% R0402

C244 10uF 10V.X5R C0402

U32 VIN VOUT CT QD ON

TPS22918DBVR SOT23-6

Sensor_3V3

C245 220pF 50V.NPO C0402

C406 10uF 10V.X5R C0402 DNP

C391 100nF 16V.X7R C0402

Design Note:

V_{IH} min = 1V.

RGB LED

Sensor_3V3

R276 1K 5% R0402

Q1 2SK3018 SOT-323-3

R282 100K 1% R0402

R277 1K 5% R0402

Q2 2SK3018 SOT-323-3

R283 100K 1% R0402

R278 2.2K 1% R0402

Q3 2SK3018 SOT-323-3

R284 100K 1% R0402

D25 RGB LED_4P_1_6X1_6MM_RGB

Passive Buzzer

VCC_3V3_MAIN VDD_3V3

R616 0R 5% DNP

R617 0R 5% DNP

VCC_3V3_IMU

[11] F22/GPIO0_29/PWM/3V3

R279 1K 5% R0402

Q4 BC817W SOT323

R281 10K 1% R0402

SMD Buzzer 3.6V Vop=2.5-4.5V:4P-8.5*8.5*3.0mm buzzer4p_7d1_8_5x8_5x3mm

IMU Sensor

VCC_3V3_MAIN

R287 0R 5% IMU_SCL

R289 0R 5% IMU_SDA

VCC_3V3_IMU

R290 0R 5% DNP

R291 0R 5% DNP

R292 0R 5% DNP

R294 0R 5% DNP

R295 100K 1% R0402

[22] IMU_INT/3V3

U33 SDO/SA0 SCL SDA SDx SCx INT1 VDDIO VDD GND1 GND2

LSM6DS3TR-C LGA14_0d5_2_5x3x0_86mm

VCC_3V3_IMU

C248 100nF 16V.X7R C0402

C249 100nF 16V.X7R C0402

Software: Default: High INT: low

T & H Sensor

[5,15,19,21] I2C0_SDA/3V3

[5,15,19,21] I2C0_SCL/3V3

R326 0R 5% R0402

R325 0R 5% R0402

U40 SDA VDD SCL VSS GND_EP

SHT40 dfn4_0d8_1_5x1_5x0_54mm_rko DNP

Sensor_3V3

R327 0R 5% R0402

C410 100nF 16V.X7R C0402

Light Sensor

Sensor_3V3

C411 100nF 16V.X7R C0402

U45 TLV9061IDPWR X2SON4_0d48_0_8x0_8x0_4mm

R511 10K 1% R0402

R513 12K 1% R0402

C282 10nF 16V.X7R C0201

U45

R573 0R 5% R0402

Sensor_3V3

Q7 ALS-PT19-315C/L177/TR8

SNR2_1d5_1_7x0_8x0_6mm

R509 100K 1% R0402

Design Note:

V_{out}(max.) = V_{cc} - 0.4V

Layout Note:

C282 Caps should be close to AM62L.

Design Note:

Input: 0 - 2V₉ to Output: 0 - 1V₆

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Title: BeagleBadge AM62L

Size: A3

Document Number: 21 Buzzer & IMU & Light

Rev: RevA

Date: Friday, November 07, 2025

Sheet: 21 of 25

Sensor Power Switch

VDD_1V8

R272 10K 1% R0402

U61 VCC

SN74LVC1G08DBVR sot23-5

VDD_1V8

C243 100nF 16V.X7R C0402

VDD_3V3

R273 10K 1% R0402

C244 10uF 10V.X5R C0402

U32 VIN VOUT CT QD ON

TPS22918DBVR SOT23-6

Sensor_3V3

C245 220pF 50V.NPO C0402

C406 10uF 10V.X5R C0402 DNP

C391 100nF 16V.X7R C0402

Design Note:

V_{IH} min = 1V.

RGB LED

Sensor_3V3

R276 1K 5% R0402

Q1 2SK3018 SOT-323-3

R282 100K 1% R0402

R277 1K 5% R0402

Q2 2SK3018 SOT-323-3

R283 100K 1% R0402

R278 2.2K 1% R0402

Q3 2SK3018 SOT-323-3

R284 100K 1% R0402

D25 RGB LED_4P_1_6X1_6MM_RGB

Passive Buzzer

VCC_3V3_MAIN VDD_3V3

R616 0R 5% DNP

R617 0R 5% DNP

VCC_3V3_IMU

[11] F22/GPIO0_29/PWM/3V3

R279 1K 5% R0402

Q4 BC817W SOT323

R281 10K 1% R0402

SMD Buzzer 3.6V Vop=2.5-4.5V:4P-8.5*8.5*3.0mm buzzer4p_7d1_8_5x8_5x3mm

IMU Sensor

VCC_3V3_MAIN

R287 0R 5% IMU_SCL

R289 0R 5% IMU_SDA

VCC_3V3_IMU

R290 0R 5% DNP

R291 0R 5% DNP

R292 0R 5% DNP

R294 0R 5% DNP

R295 100K 1% R0402

[22] IMU_INT/3V3

U33 SDO/SA0 SCL SDA SDx SCx INT1 VDDIO VDD GND1 GND2

LSM6DS3TR-C LGA14_0d5_2_5x3x0_86mm

VCC_3V3_IMU

C248 100nF 16V.X7R C0402

C249 100nF 16V.X7R C0402

Software: Default: High INT: low

T & H Sensor

[15] V23/ADC0_AIN2/1V8

R511 10K 1% R0402

R513 12K 1% R0402

C282 10nF 16V.X7R C0201

U40 SDA VDD SCL VSS GND_EP

SHT40 dfn4_0d8_1_5x1_5x0_54mm_rko DNP

R326 0R 5% R0402

R327 0R 5% R0402

R325 0R 5% R0402

Sensor_3V3

C410 100nF 16V.X7R C0402

Light Sensor

Sensor_3V3

C411 100nF 16V.X7R C0402

U45 X2SON4_0d48_0_8x0_8x0_4mm

TLV9061IDPWR

R573 0R 5% R0402

R510 0R 5% R0402

Q7 ALS-PT19-315C/L177/TR8

SNR2_1d5_1_7x0_8x0_6mm

R509 100K 1% R0402

Design Note:

V_{out}(max.) = V_{cc} - 0.4V

Layout Note:

C282 Caps should be close to AM62L.

Design Note:

Input: 0 - 2V9 to Output: 0 - 1V6

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Title: BeagleBadge AM62L

Size: A3

Document Number: 21 Buzzer & IMU & Light

Date: Friday, November 07, 2025

Rev: RevA

Sheet: 21 of 25

Sensor Power Switch

VDD_1V8

R272 10K 1% R0402

U61 VCC

SN74LVC1G08DBVR sot23-5

VDD_1V8

C243 100nF 16V.X7R C0402

VDD_3V3

R273 10K 1% R0402

C244 10uF 10V.X5R C0402

U32 VIN VOUT CT QD ON

TPS22918DBVR SOT23-6

Sensor_3V3

C245 220pF 50V.NPO C0402

C406 10uF 10V.X5R C0402 DNP

C391 100nF 16V.X7R C0402

Design Note:

V_{IH} min = 1V.

RGB LED

Sensor_3V3

R276 1K 5% R0402

Q1 2SK3018 SOT-323-3

R282 100K 1% R0402

R277 1K 5% R0402

Q2 2SK3018 SOT-323-3

R283 100K 1% R0402

R278 2.2K 1% R0402

Q3 2SK3018 SOT-323-3

R284 100K 1% R0402

D25 RGB LED_4P_1_6X1_6MM_RGB

Passive Buzzer

VCC_3V3_MAIN VDD_3V3

R616 0R 5% DNP

R617 0R 5% DNP

VCC_3V3_IMU

[11] F22/GPIO0_29/PWM/3V3

R279 1K 5% R0402

Q4 BC817W SOT323

R281 10K 1% R0402

SMD Buzzer 3.6V Vop=2.5-4.5V:4P-8.5*8.5*3.0mm buzzer4p_7d1_8_5x8_5x3mm

IMU Sensor

VCC_3V3_MAIN

R287 0R 5% IMU_SCL

R289 0R 5% IMU_SDA

VCC_3V3_IMU

R290 0R 5% DNP

R291 0R 5% DNP

R292 0R 5% DNP

R294 0R 5% DNP

R295 100K 1% R0402

[22] IMU_INT/3V3

U33 SDO/SA0 SCL SDA SDx SCx INT1 VDDIO VDD GND1 GND2

LSM6DS3TR-C LGA14_0d5_2_5x3x0_86mm

VCC_3V3_IMU

C248 100nF 16V.X7R C0402

C249 100nF 16V.X7R C0402

Software: Default: High INT: low

T & H Sensor

[15] V23/ADC0_AIN2/1V8

R511 10K 1% R0402

R513 12K 1% R0402

C282 10nF 16V.X7R C0201

U40 SDA VDD SCL VSS GND_EP

SHT40 dfn4_0d8_1_5x1_5x0_54mm_rko DNP

R326 0R 5% R0402

R327 0R 5% R0402

R325 0R 5% R0402

Sensor_3V3

C410 100nF 16V.X7R C0402

Light Sensor

Sensor_3V3

C411 100nF 16V.X7R C0402

U45 X2SON4_0d48_0_8x0_8x0_4mm

TLV9061IDPWR

R573 0R 5% R0402

R510 0R 5% R0402

Q7 ALS-PT19-315C/L177/TR8

SNR2_1d5_1_7x0_8x0_6mm

R509 100K 1% R0402

Design Note:

V_{out}(max.) = V_{cc} - 0.4V

Layout Note:

C282 Caps should be close to AM62L.

Design Note:

Input: 0 - 2V₉ to Output: 0 - 1V₆

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Title: BeagleBadge AM62L

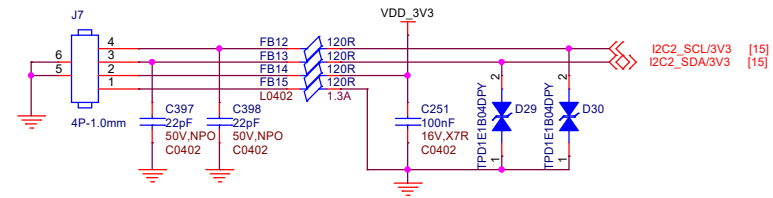
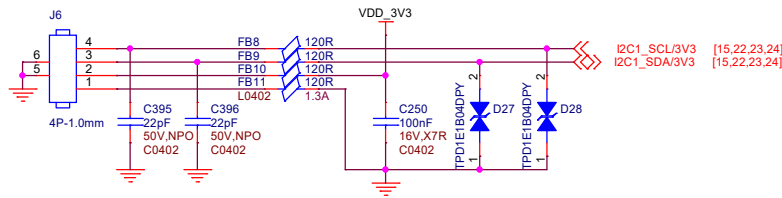
Size: A3

Document Number: 21 Buzzer & IMU & Light

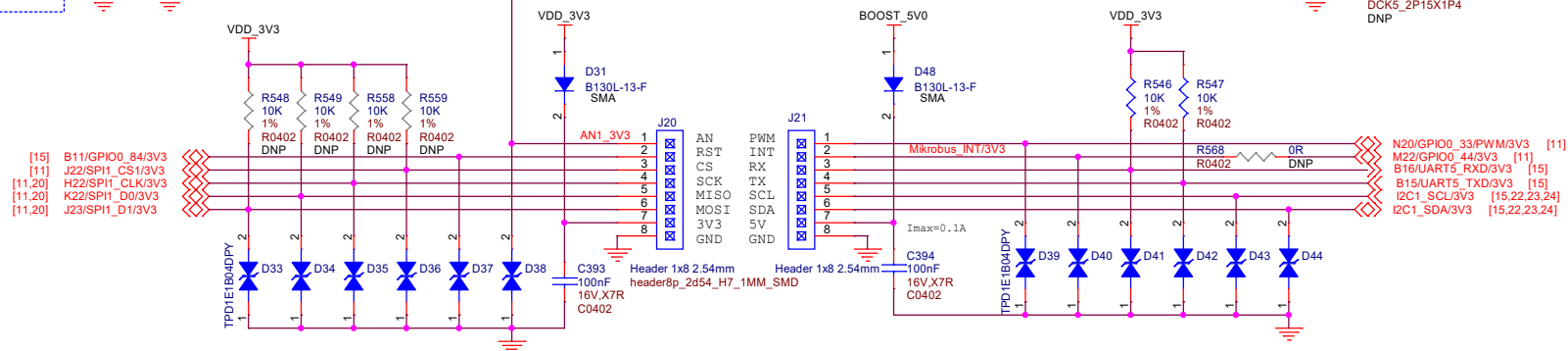
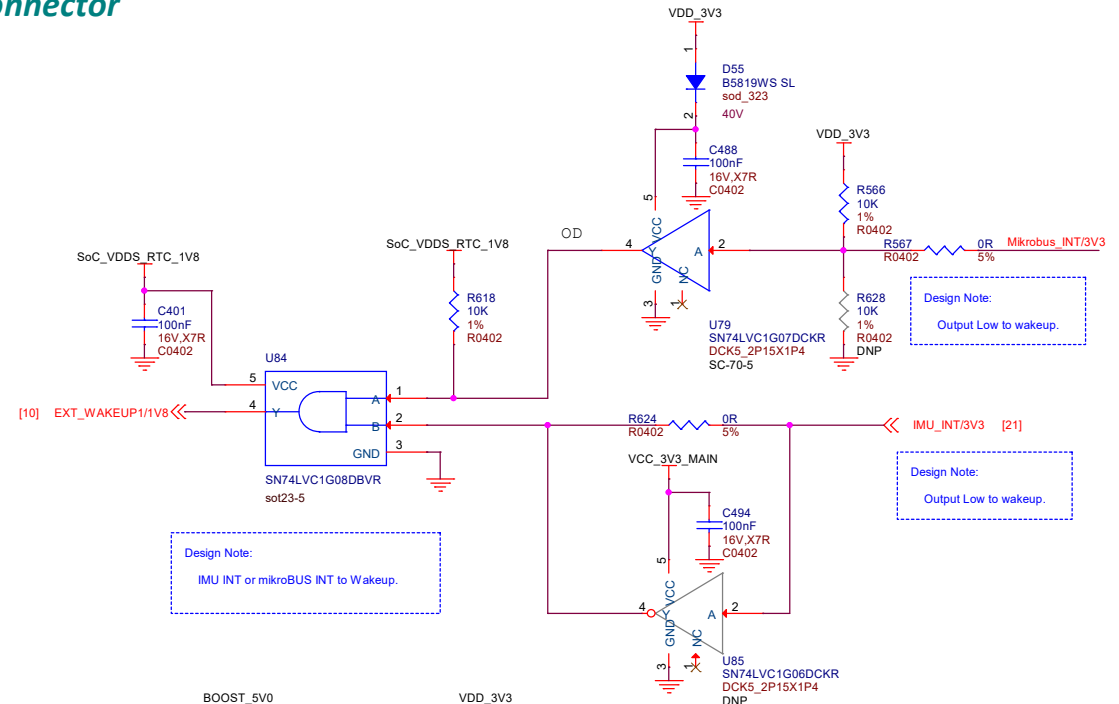
Date: Friday, November 07, 2025

Rev: RevA

Sheet: 21 of 25



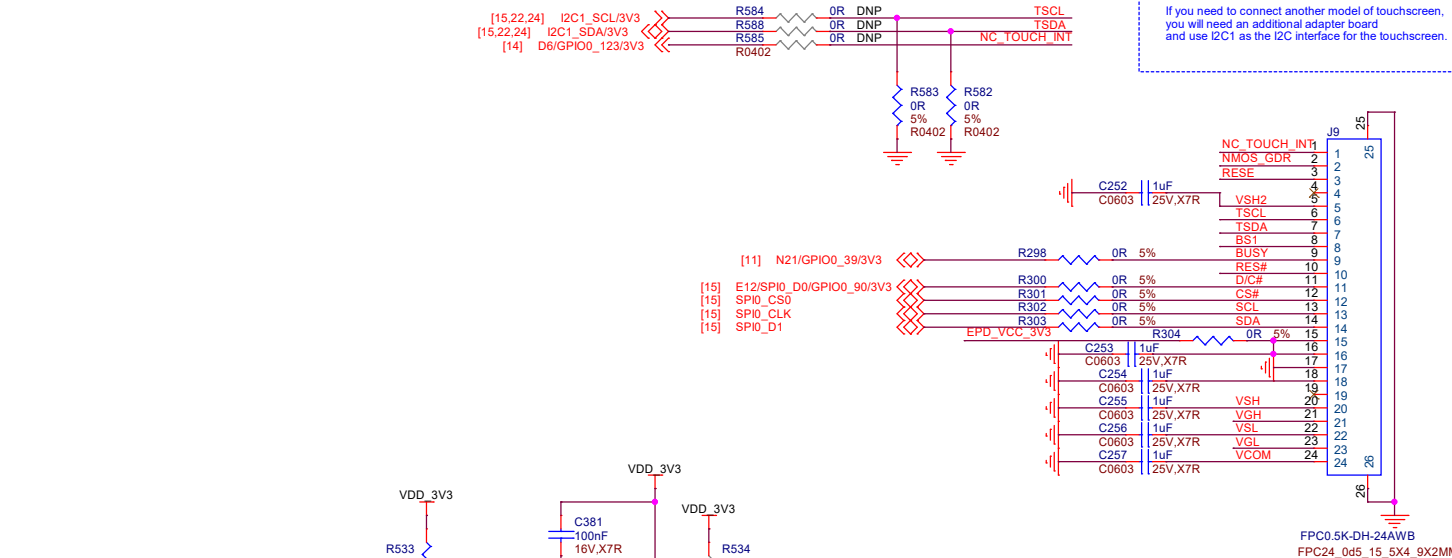
QWIIC Connector



MikroBus

Layout Note:
Close to epaper connector

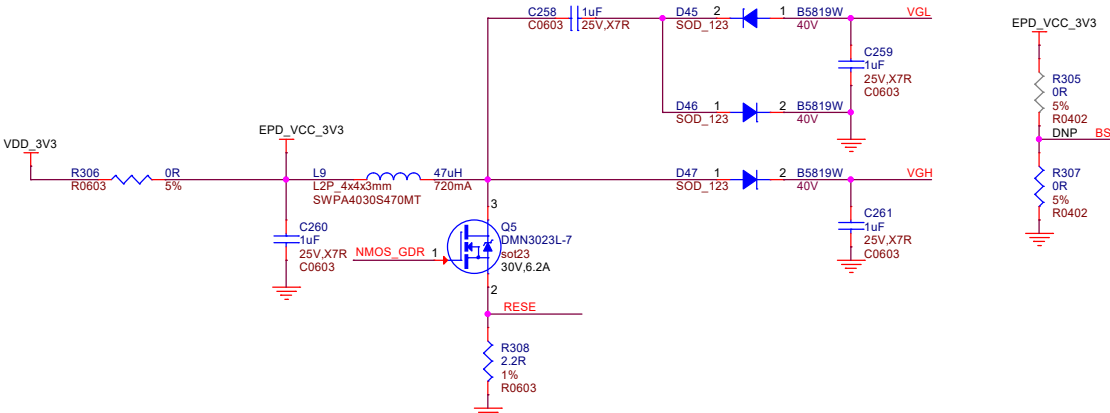
Design Note:
If you need to connect another model of touchscreen, you will need an additional adapter board and use I2C1 as the I2C interface for the touchscreen.



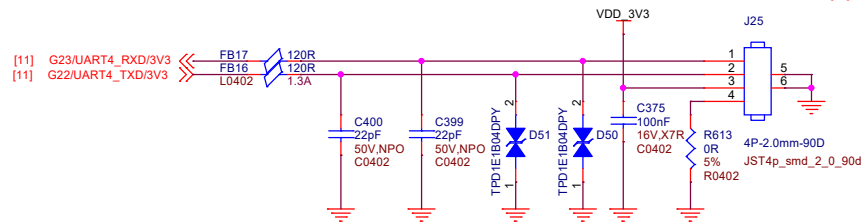
Input/Output Terminals

Pin #	Signal	No.
1	NC	
2	GBR	
3	RESE	
4	NC	
5	VSH2	
6	TSCL	
7	TSDA	
8	BS1	
9	BUSY	
10	RES#	
11	D/C#	
12	CS#	
13	SCL	
14	SDA	
15	NC	
16	NC	
17	VDDIO	
18	VCI	
19	VSS	
20	VDD	
21	VPP	
22	VSH1	
23	VGH	
24	VGL	
25	VCOM	

ePaper Connector

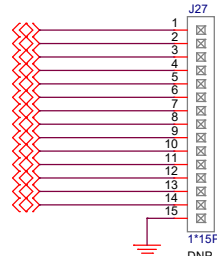


Design Note:
BS1 = 0, 4-Line SPI Mode
BS1 = 1, 3-Line SPI Mode

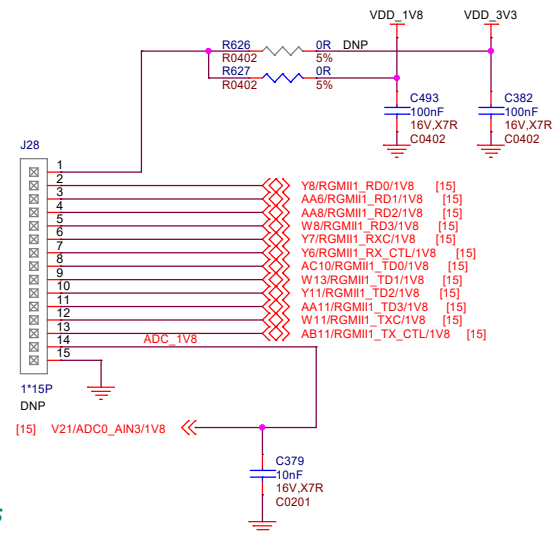


GROVE Connector

- [15] AB9/RGMII2_RD0/1V8
- [15] AC9/RGMII2_RD1/1V8
- [15] AB10/RGMII2_RD2/1V8
- [15] AB8/RGMII2_RD3/1V8
- [15] AC7/RGMII2_RXC/1V8
- [15] AC8/RGMII1_RX_CTL/1V8
- [15] AC12/RGMII2_TD0/1V8
- [15] AB13/RGMII2_TD1/1V8
- [15] AA12/RGMII2_TD2/1V8
- [15] AA13/RGMII2_TD3/1V8
- [15] Y13/RGMII2_TXC/1V8
- [15] AB12/RGMII2_TX_CTL/1V8
- [10] AC13/MDIO0_MDIO/1V8
- [10] AC15/MDIO0_MDC/1V8

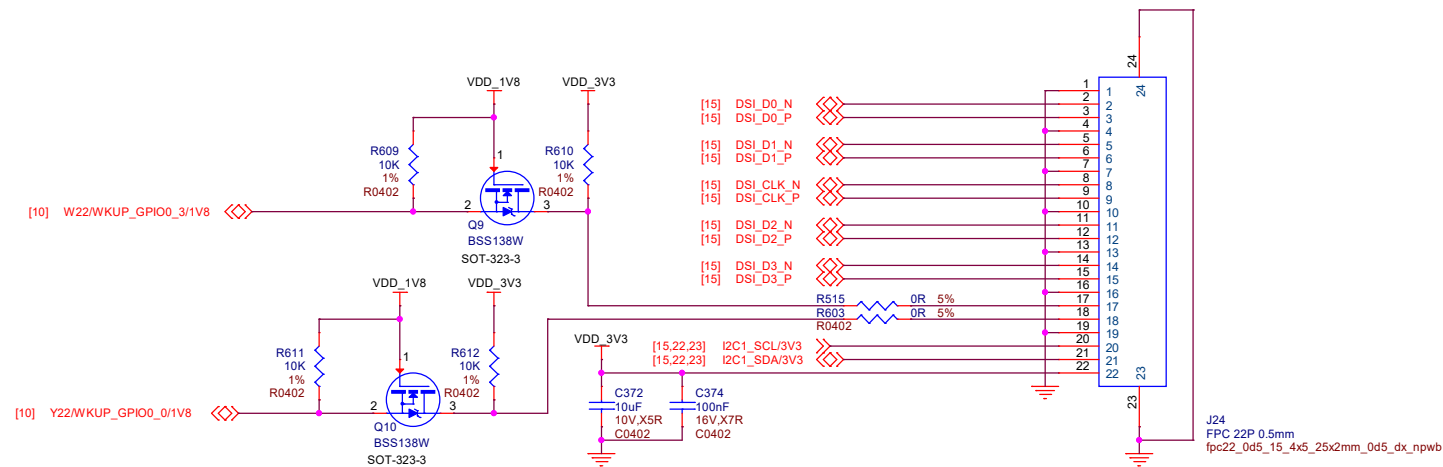


Other GPIOs



Layout Note:

C379 Caps should be close to AM62L.



DSI Connector

